

Primal-Dual Prediction-Correction Method with Tunable Memory for Linearly Constrained Time-Varying Convex Optimization*

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Abstract This paper presents a primal-dual prediction-correction (PD-PC) method for solving linearly constrained time-varying convex optimization problems, which frequently arise in control, signal processing, and online learning applications. The proposed method establishes a novel integration of primal-dual gradient dynamics with a discrete-time prediction-correction structure, specifically designed for problems with time-dependent linear constraints. A tunable memory parameter is introduced in the prediction phase to perform linear extrapolation using past iterates, enabling a flexible trade-off between the amount of historical information stored and the computational cost of correction. In the correction phase, primal and dual variables are updated via gradient descent-ascent iterations, thus maintaining the computational efficiency of a first-order method without requiring Hessian or high-order derivative computations. Theoretical analysis shows that the method achieves $\mathcal{O}(h^2)$ asymptotic tracking accuracy for both primal and dual variables, matching the state-of-the-art performance among first-order methods even in unconstrained settings. Numerical experiments on problems with both time-invariant and time-varying constraints validate the theoretical findings and demonstrate the method's effectiveness.

Keywords Convex optimization, prediction-correction method, primal-dual method, time-varying optimization, tunable memory.

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1 Introduction

Many modern systems, from control and robotics to signal processing and learning, rely on solving optimization problems whose objectives or constraints evolve over time. This time-varying nature reflects evolving measurements, environments, or network states, and is central to applications such as adaptive filtering and target tracking^[1–3], model predictive control^[4–6], resource allocation in communication networks^[7, 8], online learning^[9–12], and distributed optimization^[13–15]. Unlike time-invariant settings where the solution remains fixed, time-varying optimization requires tracking a continuously shifting optimum, often under limited sampling or computational resources. Formally, we consider a linearly constrained time-varying convex problem

$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}, t) \quad \text{s.t.} \quad \mathbf{A}(t)\mathbf{x} = \mathbf{b}(t), \quad (1)$$

where $f : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}$ is a function such that, for each fixed $t \geq 0$, the mapping $\mathbf{x} \mapsto f(\mathbf{x}, t)$ is convex and twice differentiable, and $\mathbf{A} : \mathbb{R}_+ \rightarrow \mathbb{R}^{m \times d}$, $\mathbf{b} : \mathbb{R}_+ \rightarrow \mathbb{R}^m$. For each $\mathbf{x} \in \mathbb{R}^d$, the mappings $t \mapsto f(\mathbf{x}, t)$ and $t \mapsto (\mathbf{A}(t), \mathbf{b}(t))$ are continuously differentiable, implying that the problem data evolve continuously over time. More specific conditions on $f(\mathbf{x}, t)$, $\mathbf{A}(t)$, and $\mathbf{b}(t)$, including strong convexity, Lipschitz smoothness, and temporal regularity, will be detailed in Section 2.

Problem (1) encompasses a broad class of models encountered across various application domains. One prominent instance arises in signal processing, where the goal is to track a time-varying reference signal. For example, by setting $f(\mathbf{x}, t) = \frac{1}{2}\|\mathbf{x} - \mathbf{r}(t)\|^2$, with $\mathbf{r}(t)$ denoting the target signal at time t , the optimal solution of the unconstrained problem coincides with $\mathbf{r}(t)$. The equality constraint $\mathbf{A}(t)\mathbf{x} = \mathbf{b}(t)$ can encode physical laws, sensor consistency, or calibration requirements imposed by system dynamics and measurement models^[16–18]. Another representative example originates from consensus problems in control and distributed optimization. When $\mathbf{x}$ stacks local agent variables and $\mathbf{A}(t) = \mathcal{L}(t)$ is a graph Laplacian with $\mathbf{b}(t) = \mathbf{0}$, the constraint enforces agreement among agents, thereby modeling a time-varying consensus-based optimization problem^[19–21]. These two cases illustrate the structural diversity and practical significance of Problem (1), motivating the development of both continuous-time and discrete-time algorithms for solving such problems.

To solve Problem (1), a class of continuous-time algorithms has been developed based on primal-dual dynamics. Representative works such as [22, 23] constructed gradient-flow or projection-based saddle-point systems that steer the primal-dual variables toward the optimal trajectory and established exponential convergence to a bounded neighborhood under some regularity conditions. An alternative approach employed prediction-correction dynamics^[24], where a prediction term accounts for the time drift of the optimizer and a correction term enforces instantaneous optimality. Such methods achieve exponential convergence to the time-varying solution trajectory from arbitrary initial conditions. Extensions to distributed implementations have been proposed in [25, 26], where consensus-based mechanisms are incorporated into prediction-correction dynamics for distributed optimization problems with time-varying constraints, yielding provable exponential tracking convergence. Despite their analytical tractabil-

ity and theoretical guarantees, however, these continuous-time algorithms are often impractical for direct deployment on digital platforms such as sensors and actuators.

To address this issue, a discrete-time framework is typically adopted by sampling Problem (1) at sampling instants $\{t_k\}_{k \in \mathbb{N}}$ with fixed sampling period $h := t_k - t_{k-1}$. This discretization yields a sequence of time-invariant optimization problems:

$$\mathbf{x}^*(t_k) := \arg \min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}, t_k) \quad \text{s.t.} \quad \mathbf{A}(t_k)\mathbf{x} = \mathbf{b}(t_k). \quad (2)$$

The goal in solving Problem (2) is to generate a discrete sequence $\{\mathbf{x}_k\}$ that accurately tracks the optimal trajectory $\{\mathbf{x}^*(t_k)\}$. The tracking performance is commonly evaluated via the *asymptotic primal tracking error* (APTE), defined as $\limsup_{k \rightarrow \infty} \|\mathbf{x}_k - \mathbf{x}^*(t_k)\|$. In the unconstrained setting, the analogous metric is the *asymptotic tracking error* (ATE)^[27]. The accuracy of a discrete-time algorithm is characterized by the asymptotic order of such errors with respect to the sampling period h . For example, $\mathcal{O}(h^2)$ APTE indicates that the algorithm achieves second-order tracking accuracy.

Several discrete-time algorithms have been developed to solve Problem (2). The primal-dual gradient method in [28], originally developed for time-invariant linearly constrained problems, achieves at most $\mathcal{O}(h)$ accuracy when applied to time-varying programs. To improve tracking accuracy, a class of prediction-correction inspired algorithms—similar to their continuous-time counterparts—has been explored in the discrete-time setting. Specifically, [29] incorporated a projected gradient-based correction, while [30] proposed a dual ascent-based scheme. Under time-invariant linear constraints and regularity assumptions, both methods attain $\mathcal{O}(h^2)$ accuracy when sufficient prediction steps are taken. Extending to distributed optimization, [31] integrated a Newton-type correction into a distributed prediction-correction framework, guaranteeing at least $\mathcal{O}(h^2)$ APTE and reaching $\mathcal{O}(h^4)$ accuracy under stronger assumptions on smoothness and initialization. However, this improvement comes at the cost of increased computational complexity due to the requirement for Hessian evaluations.

In this paper, we propose a novel primal-dual prediction-correction method, termed **PD-PC**, which achieves $\mathcal{O}(h^2)$ tracking accuracy for Problem (2) with time-varying linear constraints. Table 1 compares PD-PC with representative discrete-time algorithms, highlighting our method's advantages in constraint generality and tracking accuracy.

Table 1 Comparison of discrete-time methods for linearly constrained time-varying convex optimization

Method	Implementation	Constraint	Tracking Accuracy (APTE)
[28] PD	Centralized	$\mathbf{A}$ invariant, $\mathbf{b}$ invariant	$\mathcal{O}(h)$
[29] C-FOPC	Centralized	$\mathbf{A}$ invariant, $\mathbf{b}$ invariant	$\mathcal{O}(h) \rightarrow \mathcal{O}(h^2)$ (sufficient predictions)
[30] Dual-PC	Centralized	$\mathbf{A}$ invariant, $\mathbf{b}$ invariant	$\mathcal{O}(h) \rightarrow \mathcal{O}(h^2)$ (sufficient predictions)
[31] Newton-PC	Distributed	$\mathbf{A}$ invariant, $\mathbf{b}$ varying	$\mathcal{O}(h^2) \rightarrow \mathcal{O}(h^4)$ (stronger regularity)
Our work: PD-PC	Centralized	$\mathbf{A}$ varying , $\mathbf{b}$ varying	$\mathcal{O}(h^2)$

Main Contributions Compared with previous work, the main contributions of our study are summarized as follows.

- 1) We propose a discrete-time method for linearly constrained time-varying convex optimization with general time-dependent constraint coefficient $\mathbf{A}(t)$ and $\mathbf{b}(t)$, beyond the time-invariant or partially time-varying settings in [22, 23, 30, 31]. This generality is nontrivial: Differentiating the KKT conditions with time-varying constraints produces extra terms involving $\dot{\mathbf{A}}(t)$ and $\dot{\mathbf{b}}(t)$ (and higher-order variations). Controlling these terms is essential for deriving time-uniform bounds along the optimal trajectory and, consequently, rigorous tracking-error guarantees.
- 2) The proposed method integrates the primal-dual gradient method with the prediction-correction framework, achieving a high tracking accuracy of $\mathcal{O}(h^2)$ with only a single primal and dual gradient update in each correction step. In contrast, the Dual-PC method^[30] relies on multiple prediction steps and requires solving an inner optimization problem at each correction step, yet does not ensure comparable accuracy. While higher-order methods such as [31, 32] can achieve $\mathcal{O}(h^4)$ accuracy, they incur significant computational overhead from Hessians and high-order mixed partial derivatives.
- 3) We introduce a novel prediction mechanism that incorporates a tunable memory parameter, enabling linear extrapolation using past iterates. This design establishes a controllable trade-off between the amount of historical information stored (memory cost) and the number of correction steps required (computational cost), offering a flexible balance between accuracy and efficiency.

Paper Organization The remainder of the paper is organized as follows. Section 2 formalizes the linearly constrained time-varying problem, and states the standard assumptions used throughout the paper. Section 3 presents the proposed PD-PC method, detailing its memory- r prediction and τ -step primal-dual correction. Section 4 develops the theoretical guarantees. Section 5 provides numerical results under various constraint settings to validate the efficacy of the proposed method.

Notations Let $\mathbb{R}$ denote the set of real numbers, $\mathbb{R}_+$ denote the set of nonnegative reals, and $\mathbb{N}$ denote the set of positive integers. For any vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, the inner product is defined as $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y}$, and $\|\mathbf{x}\|$ denotes the Euclidean norm, which also coincides with the spectral norm when applied to a matrix. The symbols $\mathbf{0}$ and $\mathbf{I}$ denote the zero vector and the identity matrix, respectively. For a matrix $\mathbf{A}$, its inverse and Moore-Penrose pseudoinverse are denoted by $\mathbf{A}^{-1}$ and $\mathbf{A}^\dagger$; its rank, null space, and column space (image) are denoted by $\text{rank}(\mathbf{A})$, $\ker(\mathbf{A})$, and $\text{im}(\mathbf{A})$. The quantities $\sigma_{\max}(\mathbf{A})$ and $\sigma_{\min}^+(\mathbf{A})$ denote the largest and smallest nonzero singular values of $\mathbf{A}$.

Consider a smooth function $f : \mathbb{R}^d \times \mathbb{R}_+ \rightarrow \mathbb{R}$. The gradient, Hessian, and third-order derivatives of f with respect to $\mathbf{x} \in \mathbb{R}^d$ are denoted by $\nabla_{\mathbf{x}} f(\mathbf{x}, t)$, $\nabla_{\mathbf{x}\mathbf{x}} f(\mathbf{x}, t)$, and $\nabla_{\mathbf{x}\mathbf{x}\mathbf{x}} f(\mathbf{x}, t)$, respectively. The mixed derivatives $\nabla_{\mathbf{x}t} f(\mathbf{x}, t)$, $\nabla_{\mathbf{x}xt} f(\mathbf{x}, t)$, and $\nabla_{\mathbf{x}tt} f(\mathbf{x}, t)$ denote the derivatives of $\nabla_{\mathbf{x}} f(\mathbf{x}, t)$ with respect to t , of $\nabla_{\mathbf{x}\mathbf{x}} f(\mathbf{x}, t)$ with respect to t , and of $\nabla_{\mathbf{x}} f(\mathbf{x}, t)$ with respect to t twice, respectively. For any continuously differentiable function $\mathbf{x}(t) : \mathbb{R}_+ \rightarrow \mathbb{R}^d$, $\dot{\mathbf{x}}(t)$ and $\ddot{\mathbf{x}}(t)$ denote its first and second time derivatives. The same notation applies element-

wise to matrix- or vector-valued functions such as $\mathbf{A}(t)$ and $\mathbf{b}(t)$, yielding $\dot{\mathbf{A}}(t)$, $\ddot{\mathbf{A}}(t)$, $\dot{\mathbf{b}}(t)$, and $\ddot{\mathbf{b}}(t)$.

2 Problem Formulation and Assumptions

For the time-varying linearly constrained convex problem (1), we define the Lagrangian at time t as

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, t) = f(\mathbf{x}, t) + \boldsymbol{\lambda}^T (\mathbf{A}(t)\mathbf{x} - \mathbf{b}(t)), \quad (3)$$

where $\boldsymbol{\lambda} \in \mathbb{R}^m$ denotes the dual variable. Let the aggregate variable $\mathbf{z} := [\mathbf{x}^T, \boldsymbol{\lambda}^T]^T \in \mathbb{R}^{d+m}$ collect the primal and dual variables.

A pair $\mathbf{z}^*(t) = [\mathbf{x}^*(t)^T, \boldsymbol{\lambda}^*(t)^T]^T$ is called a *KKT pair* of (1) at time t if it satisfies the first-order optimality condition

$$\nabla_{\mathbf{z}} \mathcal{L}(\mathbf{z}^*(t), t) = \mathbf{0}, \quad (4)$$

which expands to the following KKT relations:

$$\nabla_{\mathbf{x}} f(\mathbf{x}^*(t), t) + \mathbf{A}(t)^T \boldsymbol{\lambda}^*(t) = \mathbf{0}, \quad \mathbf{A}(t)\mathbf{x}^*(t) = \mathbf{b}(t). \quad (5)$$

Our objective is to design a discrete-time algorithm, executed at the sampling instants t_k with sampling period h , that generates iterates $\mathbf{z}_k$ to track the sampled *primal-dual solution trajectory* $\{\mathbf{z}^*(t_k)\}$, where $\mathbf{z}^*(t)$ is defined by (4)–(5). Since discrete-time implementations generally converge to an h -dependent neighborhood of $\mathbf{z}^*(t_k)$, we aim to achieve second-order tracking accuracy, namely $\limsup_{k \rightarrow \infty} \|\mathbf{z}_k - \mathbf{z}^*(t_k)\| = O(h^2)$. In particular, this corresponds to an $O(h^2)$ APTE, i.e., $\limsup_{k \rightarrow \infty} \|\mathbf{x}_k - \mathbf{x}^*(t_k)\| = O(h^2)$.

To ensure the well-posedness of (1), we introduce the following assumptions. These assumptions are standard in the analysis of prediction-correction methods for time-varying optimization problems^[22–24, 26, 28–32].

Assumption 2.1 For all $t \geq 0$, the matrix $\mathbf{A}(t) \in \mathbb{R}^{m \times d}$ has full row rank and its nonzero singular values are uniformly bounded away from 0 and ∞ ; namely,

$$\text{rank}(\mathbf{A}(t)) = m, \quad \sigma_{\min}^+(\mathbf{A}(t)) \geq \underline{\sigma} > 0, \quad \sigma_{\max}(\mathbf{A}(t)) \leq \bar{\sigma} < \infty, \quad \forall t \geq 0. \quad (6)$$

This assumption guarantees feasibility of (1) (since $\text{im}(\mathbf{A}(t)) = \mathbb{R}^m$ when $\mathbf{A}(t)$ has full row rank) and, for convex equality-constrained problems, strong duality. Hence at each t a KKT pair exists and its primal component $\mathbf{x}^*(t)$ solves Problem (1). Moreover, the two-sided spectral bounds provide uniform control of the nonzero singular values of $\mathbf{A}(t)$ over time: $\kappa^+(\mathbf{A}(t)) := \sigma_{\max}(\mathbf{A}(t))/\sigma_{\min}^+(\mathbf{A}(t)) \leq \bar{\sigma}/\underline{\sigma}$. Hence both $\|\mathbf{A}(t)\| = \sigma_{\max}(\mathbf{A}(t))$ and $\|\mathbf{A}^\dagger(t)\| = 1/\sigma_{\min}^+(\mathbf{A}(t))$ are uniformly bounded. This excludes regimes with unbounded amplification ($\sigma_{\max} \rightarrow \infty$) or near-singularity on the row space ($\sigma_{\min}^+ \rightarrow 0$), and yields time-uniform admissible step-size intervals for the primal-dual gradient updates in our proposed method.

Assumption 2.2 For all $t \geq 0$, the function $f(\cdot, t) : \mathbb{R}^d \rightarrow \mathbb{R}$ is twice differentiable in $\mathbf{x}$ and satisfies

$$\nu \mathbf{I} \preceq \nabla_{\mathbf{x}\mathbf{x}} f(\mathbf{x}, t) \preceq \delta \mathbf{I}, \quad \forall \mathbf{x} \in \mathbb{R}^d, \quad \forall t \geq 0, \quad (7)$$

for some constants $0 < \nu \leq \delta < \infty$. That is, $f(\cdot, t)$ is uniformly ν -strongly convex and δ -smooth (i.e., $\nabla_{\mathbf{x}} f(\mathbf{x}, t)$ is δ -Lipschitz continuous) with respect to $\mathbf{x}$.

Under Assumptions 2.1–2.2, the Hessian of the Lagrangian with respect to the aggregate variable $\mathbf{z}$,

$$\nabla_{\mathbf{z}\mathbf{z}} \mathcal{L}(\mathbf{z}^*(t), t) = \begin{bmatrix} \nabla_{\mathbf{x}\mathbf{x}} f(\mathbf{x}^*(t), t) & \mathbf{A}(t)^T \\ \mathbf{A}(t) & \mathbf{0} \end{bmatrix} \quad (8)$$

is nonsingular for every t . Differentiating the first-order optimality condition $\nabla_{\mathbf{z}} \mathcal{L}(\mathbf{z}^*(t), t) = \mathbf{0}$ with respect to t yields the differential system in [24]

$$\dot{\mathbf{z}}^*(t) = -\left(\nabla_{\mathbf{z}\mathbf{z}} \mathcal{L}(\mathbf{z}^*(t), t)\right)^{-1} \nabla_{\mathbf{z}t} \mathcal{L}(\mathbf{z}^*(t), t). \quad (9)$$

Hence, the (continuous-time) primal-dual solution trajectory $\{\mathbf{z}^*(t)\}$ is uniquely determined once $\mathbf{z}^*(0)$ is specified, and so are its components $\mathbf{x}^*(t)$ and $\boldsymbol{\lambda}^*(t)$.

To achieve high-accuracy discrete-time tracking of the time-varying KKT pair, it is further necessary to control both the magnitude of $\mathbf{z}^*(t)$ and its temporal variability. We therefore introduce the following temporal regularity assumption.

Assumption 2.3 The following quantities are bounded uniformly over $t \geq 0$:

(A1) Boundedness of the primal-dual solution trajectory:

$$\sup_{t \geq 0} \|\mathbf{x}^*(t)\| < \infty, \quad \sup_{t \geq 0} \|\boldsymbol{\lambda}^*(t)\| < \infty. \quad (10)$$

(A2) Bounded first-order temporal variations:

$$\sup_{t \geq 0} \|\nabla_{\mathbf{x}t} f(\mathbf{x}^*(t), t)\| < \infty, \quad \sup_{t \geq 0} \|\dot{\mathbf{A}}(t)\| < \infty, \quad \sup_{t \geq 0} \|\dot{\mathbf{b}}(t)\| < \infty. \quad (11)$$

(A3) Bounded higher-order temporal variations:

$$\begin{aligned} \sup_{t \geq 0} \|\nabla_{\mathbf{x}xt} f(\mathbf{x}^*(t), t)\| < \infty, \quad \sup_{t \geq 0} \|\nabla_{\mathbf{x}tt} f(\mathbf{x}^*(t), t)\| < \infty, \\ \sup_{t \geq 0} \|\nabla_{\mathbf{x}\mathbf{x}t} f(\mathbf{x}^*(t), t)\| < \infty, \quad \sup_{t \geq 0} \|\ddot{\mathbf{A}}(t)\| < \infty, \quad \sup_{t \geq 0} \|\ddot{\mathbf{b}}(t)\| < \infty. \end{aligned} \quad (12)$$

Proposition 2.4 Suppose Assumptions 2.1–2.3 hold. Then the unique primal-dual solution trajectory $\mathbf{z}^*(t) = [\mathbf{x}^*(t)^T, \boldsymbol{\lambda}^*(t)^T]^T$ satisfying the differential system (9) is continuously differentiable for $t \geq 0$. Moreover, its second time derivative exists for all $t \geq 0$, and the following uniform bounds hold:

$$\sup_{t \geq 0} \|\dot{\mathbf{z}}^*(t)\| < \infty, \quad \sup_{t \geq 0} \|\ddot{\mathbf{z}}^*(t)\| < \infty. \quad (13)$$

Proof The proof of Proposition 2.4 can be found in Appendix A. ■

Remark 2.5 The full-row-rank requirement in Assumption 2.1 (i.e., $\text{rank}(\mathbf{A}(t)) = m$) can be relaxed to a *uniform constant-rank condition* (a weaker form of the *Linear Independence Constraint Qualification*, LICQ), under which $\text{rank}(\mathbf{A}(t))$ remains constant over time even if

$\mathbf{A}(t)$ is not full-row-rank. Under this condition, the dual variable $\boldsymbol{\lambda}^*(t)$ may not be unique; however, its projection onto $\text{im}(\mathbf{A}(t))$ is uniquely defined. Since our derivations involve only $\mathbf{A}(t)^T \boldsymbol{\lambda}^*(t)$ and the feasibility residual $\mathbf{A}(t)\mathbf{x} - \mathbf{b}(t)$, the subsequent analysis can be extended to the constant-rank setting with minor modifications; see [28], Lemma 2, for a related discussion. For clarity, we focus on the full-row-rank case in the main text.

Remark 2.6 Assumption 2.2 is stated in terms of strong convexity for clarity, but the analysis remains valid when $f(\cdot, t)$ is convex, by considering a regularized function $f_\rho(\mathbf{x}, t) = f(\mathbf{x}, t) + \frac{\rho}{2} \|\mathbf{A}(t)\mathbf{x} - \mathbf{b}(t)\|^2$ with $\rho > 0$. Its Hessian takes the form $\nabla_{\mathbf{x}\mathbf{x}} f_\rho = \nabla_{\mathbf{x}\mathbf{x}} f + \rho \mathbf{A}^T \mathbf{A}$, which is uniformly positive definite under Assumption 2.1 (i.e., $\sigma_{\min}^+(\mathbf{A}(t)) \geq \underline{\sigma} > 0$) and the condition $\ker(\nabla_{\mathbf{x}\mathbf{x}} f) \cap \ker(\mathbf{A}) = \{\mathbf{0}\}$. Thus, $f_\rho(\cdot, t)$ becomes uniformly strongly convex with modulus at least $\rho \sigma_{\min}^+(\mathbf{A}(t))^2$. Since the augmentation vanishes on the feasible set $\{\mathbf{x} : \mathbf{A}(t)\mathbf{x} = \mathbf{b}(t)\}$, the primal minimizer and KKT pairs are unchanged, and all results remain valid upon replacing f with f_ρ . The same argument applies under the LICQ condition discussed above. Therefore, while our analysis is framed within the context of strong convexity, it also applies to general convex objectives.

Remark 2.7 Assumption 2.3 concerns the temporal regularity of the optimal primal-dual solution trajectory $\mathbf{z}^*(t)$ induced by the problem structure. In particular, (A2)–(A3) are standard temporal-regularity conditions in time-varying optimization: They ensure the uniform boundedness of $\dot{\mathbf{z}}^*(t)$ and $\ddot{\mathbf{z}}^*(t)$ as shown in Proposition 2.4, and are commonly adopted in linearly constrained settings^[29–31] as well as in unconstrained settings^[32–34]. For the general time-varying constrained case, (A1) further bounds $\mathbf{z}^*(t)$ itself, which is convenient for deriving time-uniform estimates when the constraint data vary with time. Moreover, (A1) can be replaced by the existence of a reference point $\mathbf{x}_{\text{ref}} \in \mathbb{R}^d$ (e.g., $\mathbf{x}_{\text{ref}} = \mathbf{0}$) such that $\|\nabla_{\mathbf{x}} f(\mathbf{x}_{\text{ref}}, t)\|$ is uniformly bounded in t , which is often easier to verify; more details are given in Appendix B.

3 Primal-Dual Prediction-Correction Method

3.1 Algorithm Design

To track the solution trajectory of a time-varying optimization problem, a common and effective strategy is a prediction-correction scheme in discrete time. At each sampling instant t_k , the algorithm mainly performs two phases: A *prediction step*, which extrapolates the KKT pair based on historical iterates, and a *correction step*, which refines the prediction using the current problem data.

Prediction Step: Given the iterates up to time t_k , we predict the KKT pair at t_{k+1} via linear extrapolation of the past primal-dual trajectory, providing a warm start for the correction phase. Inspired by [33, 34], we employ a memory- r two-point extrapolation, i.e., the degree-one (two-point) Lagrange extrapolation based on $(t_{k-r}, \mathbf{z}_{k-r})$ and $(t_k, \mathbf{z}_k)$:

$$\mathbf{z}_{k+1|k} = \frac{r+1}{r} \mathbf{z}_k - \frac{1}{r} \mathbf{z}_{k-r}, \quad (14)$$

where the past iterates $\mathbf{z}_{k-r}$ and $\mathbf{z}_k$ are available (with $\{\mathbf{z}_{-r}, \dots, \mathbf{z}_0\}$ used for initialization). As

shown in Lemma 4.1, applying the same extrapolation coefficients to the primal-dual solution trajectory $\mathbf{z}^*(t)$ yields an $O(h^2)$ one-step prediction error, which motivates the choice of (14).

Correction Step: After prediction, a correction phase is applied to steer $\mathbf{z}_{k+1|k}$ toward $\mathbf{z}^*(t_{k+1})$. This is achieved by executing multiple steps of a primal-dual gradient method, which is well suited for equality-constrained problems. Specifically, at each time step, we perform τ correction steps, initialized at the prediction $\mathbf{z}_{k+1|k}$:

$$\widehat{\mathbf{x}}_{k+1}^{(s+1)} = \widehat{\mathbf{x}}_{k+1}^{(s)} - \eta_{\mathbf{x}} \left(\nabla_{\mathbf{x}} f(\widehat{\mathbf{x}}_{k+1}^{(s)}, t_{k+1}) + \mathbf{A}(t_{k+1})^T \widehat{\boldsymbol{\lambda}}_{k+1}^{(s)} \right), \quad (15)$$

$$\widehat{\boldsymbol{\lambda}}_{k+1}^{(s+1)} = \widehat{\boldsymbol{\lambda}}_{k+1}^{(s)} + \eta_{\boldsymbol{\lambda}} \left(\mathbf{A}(t_{k+1}) \widehat{\mathbf{x}}_{k+1}^{(s+1)} - \mathbf{b}(t_{k+1}) \right), \quad (16)$$

where $\eta_{\mathbf{x}}, \eta_{\boldsymbol{\lambda}} > 0$ are the primal and dual step sizes.

Overall, the algorithm alternates between one-step linear prediction and τ -step primal-dual correction at each time t_k . The complete procedure is outlined in Algorithm 1.

Algorithm 1 Primal-dual prediction-correction with memory- r extrapolation (**PD-PC**)

Require: Tunable memory length $r \in \mathbb{N}$, primal step size $\eta_{\mathbf{x}} > 0$, dual step size $\eta_{\boldsymbol{\lambda}} > 0$, number of correction steps $\tau \geq 1$; time-varying linear operator $\mathbf{A}(t) \in \mathbb{R}^{m \times d}$ and signal $\mathbf{b}(t) \in \mathbb{R}^m$; objective $f(\cdot, t)$ with oracle access to $\nabla_{\mathbf{x}} f(\cdot, t)$; initial history $\{\mathbf{z}_{-r}, \dots, \mathbf{z}_0\}$ where $\mathbf{z} := [\mathbf{x}^T, \boldsymbol{\lambda}^T]^T \in \mathbb{R}^{d+m}$.

Ensure: Predicted-corrected sequence $\{\mathbf{z}_k\}$ tracking the time-varying KKT pairs $\{\mathbf{z}^*(t_k)\}$.

1: **for** each iteration $k = 0, 1, \dots$ **do**

2: **Predict** [cf. (14)]:

$$\mathbf{z}_{k+1|k} = \frac{r+1}{r} \mathbf{z}_k - \frac{1}{r} \mathbf{z}_{k-r}.$$

3: Decompose $\mathbf{z}_{k+1|k}$ into blocks:

$$\mathbf{x}_{k+1|k} := (\mathbf{z}_{k+1|k})_{1:d}, \quad \boldsymbol{\lambda}_{k+1|k} := (\mathbf{z}_{k+1|k})_{d+1:d+m}.$$

4: **Evaluate problem data at t_{k+1} :** Access $\nabla_{\mathbf{x}} f(\cdot, t_{k+1})$, $\mathbf{A}(t_{k+1})$, and $\mathbf{b}(t_{k+1})$.

5: **Initialize correction:**

$$\widehat{\mathbf{x}}_{k+1}^{(0)} = \mathbf{x}_{k+1|k}, \quad \widehat{\boldsymbol{\lambda}}_{k+1}^{(0)} = \boldsymbol{\lambda}_{k+1|k}.$$

6: **for** $s = 0$ to $\tau - 1$ **do**

7: **Primal-dual gradient correction** [cf. (15)–(16)]:

$$\begin{aligned} \widehat{\mathbf{x}}_{k+1}^{(s+1)} &= \widehat{\mathbf{x}}_{k+1}^{(s)} - \eta_{\mathbf{x}} \left(\nabla_{\mathbf{x}} f(\widehat{\mathbf{x}}_{k+1}^{(s)}, t_{k+1}) + \mathbf{A}(t_{k+1})^T \widehat{\boldsymbol{\lambda}}_{k+1}^{(s)} \right), \\ \widehat{\boldsymbol{\lambda}}_{k+1}^{(s+1)} &= \widehat{\boldsymbol{\lambda}}_{k+1}^{(s)} + \eta_{\boldsymbol{\lambda}} \left(\mathbf{A}(t_{k+1}) \widehat{\mathbf{x}}_{k+1}^{(s+1)} - \mathbf{b}(t_{k+1}) \right). \end{aligned}$$

8: **end for**

9: **Update iterate:**

$$\mathbf{x}_{k+1} = \widehat{\mathbf{x}}_{k+1}^{(\tau)}, \quad \boldsymbol{\lambda}_{k+1} = \widehat{\boldsymbol{\lambda}}_{k+1}^{(\tau)}, \quad \mathbf{z}_{k+1} = [\mathbf{x}_{k+1}^T, \boldsymbol{\lambda}_{k+1}^T]^T.$$

10: **end for**

3.2 Parameter Selection

The overall behavior of the algorithm is governed by the primal-dual step sizes and the memory-correction configuration. Proper selection ensures stable updates and convergence of the primal variables toward the APTE. The guiding rules are summarized below; rigorous proofs are deferred to Section 4.

1) Step sizes. Choose the primal and dual step sizes to stabilize the correction phase:

$$0 < \eta_{\mathbf{x}} < \frac{1}{\delta}, \quad (17)$$

$$0 < \eta_{\lambda} \leq \frac{\nu}{\bar{\sigma}^2}. \quad (18)$$

These bounds yield well-conditioned primal-dual gradient updates and ensure a local contraction around the KKT pair at time t_{k+1} .

2) Memory r and correction steps τ . The parameter r determines the “memory” of extrapolation used in prediction, while τ specifies the number of correction steps applied at each iteration. Define

$$c_{\mathbf{x}} := 1 - \eta_{\mathbf{x}}\eta_{\lambda}\bar{\sigma}^2, \quad c_{\lambda} := \frac{\eta_{\mathbf{x}}}{\eta_{\lambda}}, \quad (19)$$

and the contraction factor

$$\rho := \max \left\{ 1 - \eta_{\mathbf{x}}\nu(1 - \eta_{\mathbf{x}}\delta), 1 - \eta_{\mathbf{x}}\eta_{\lambda}\bar{\sigma}^2 \right\}. \quad (20)$$

Under (17)–(18), one has $c_{\mathbf{x}}, c_{\lambda} > 0$ and $\rho \in (0, 1)$. To ensure that the prediction-correction algorithm drives the primal iterates $\{\mathbf{x}_k\}$ to the APTE, it is sufficient to require

$$\rho^{\tau} \cdot \frac{\max\{c_{\mathbf{x}}, c_{\lambda}\}}{\min\{c_{\mathbf{x}}, c_{\lambda}\}} \cdot \left(\frac{r+2}{r} \right)^2 < 1. \quad (21)$$

For fixed step sizes (hence fixed $\rho, c_{\mathbf{x}}, c_{\lambda}$), a convenient sufficient choice of τ is

$$\tau > \frac{\log \left(\left(\frac{r+2}{r} \right)^2 \cdot \frac{\max\{c_{\mathbf{x}}, c_{\lambda}\}}{\min\{c_{\mathbf{x}}, c_{\lambda}\}} \right)}{\log(1/\rho)}. \quad (22)$$

In summary, the parameters must satisfy (17)–(18) and (21) (with (22) providing a practical lower bound on τ) to ensure convergence of the PD-PC method. The next section presents rigorous justification for these choices and the resulting contraction properties.

Remark 3.1 The parameter r determines the depth of memory in the predictor, as it extrapolates from the earlier iterate $\mathbf{z}_{k-r}$. A larger r uses longer history and reduces the ratio $\left(\frac{r+2}{r} \right)^2$, which is strictly decreasing and approaches 1 as r increases. This leads to a smaller lower bound on the required correction steps τ in (22). The parameter r is thus *tunable*, allowing a trade-off: Increasing r reduces the number of correction steps needed for convergence but requires retaining more past iterates; decreasing r reduces memory usage but typically requires more correction steps.

4 Theoretical Results

This section establishes the theoretical guarantees of the proposed PD-PC method. Under Assumptions 2.1–2.3, we analyze the prediction and correction phases separately and then combine them to derive the overall convergence results.

4.1 Prediction Phase

We begin by characterizing the temporal regularity of the primal-dual solution trajectory. The following lemma shows that $\mathbf{z}^*(t_{k+1})$ can be approximated by a linear extrapolation of the past KKT pairs $\mathbf{z}^*(t_k)$ and $\mathbf{z}^*(t_{k-r})$, with an approximation error of order $\mathcal{O}(h^2)$.

Lemma 4.1 *Suppose Assumptions 2.1–2.3 hold. Then the primal-dual solution trajectory satisfies*

$$\mathbf{z}^*(t_{k+1}) = \frac{r+1}{r} \mathbf{z}^*(t_k) - \frac{1}{r} \mathbf{z}^*(t_{k-r}) + \mathcal{O}(h^2). \quad (23)$$

Proof The proof of Lemma 4.1 can be found in Appendix C. ■

Notably, the coefficients in Lemma 4.1 coincide with those used in the prediction step (14) of Algorithm 1. Combining (14) and (23) and applying the triangle inequality, yielding the prediction error bound:

$$\|\mathbf{z}_{k+1|k} - \mathbf{z}^*(t_{k+1})\| \leq \frac{r+1}{r} \|\mathbf{z}_k - \mathbf{z}^*(t_k)\| + \frac{1}{r} \|\mathbf{z}_{k-r} - \mathbf{z}^*(t_{k-r})\| + \varepsilon h^2, \quad (24)$$

where $\varepsilon > 0$ is bounded as in (C.4), which appears in the proof of Lemma 4.1 in Appendix C.

Inequality (24) bounds the prediction error at t_{k+1} by a linear combination of the past tracking errors $\|\mathbf{z}_k - \mathbf{z}^*(t_k)\|$ and $\|\mathbf{z}_{k-r} - \mathbf{z}^*(t_{k-r})\|$, plus an $\mathcal{O}(h^2)$ discretization error.

4.2 Correction Phase

We fix t_{k+1} and analyze the correction phase as a time-invariant primal-dual gradient procedure with τ iterations, initialized at the predictor $\mathbf{z}_{k+1|k}$.

We adopt the following block-weighted quadratic Lyapunov function candidate (rather than $\frac{1}{2} \|\mathbf{z}_k - \mathbf{z}^*(t_k)\|^2$), since the primal and dual updates use distinct step sizes $\eta_{\mathbf{x}}$ and $\eta_{\boldsymbol{\lambda}}$; the weights $c_{\mathbf{x}}$ and $c_{\boldsymbol{\lambda}}$ align the metric with the contraction induced by the coupled updates:

$$V_k := c_{\mathbf{x}} \|\mathbf{x}_k - \mathbf{x}^*(t_k)\|^2 + c_{\boldsymbol{\lambda}} \|\boldsymbol{\lambda}_k - \boldsymbol{\lambda}^*(t_k)\|^2, \quad (25)$$

where $c_{\mathbf{x}}$ and $c_{\boldsymbol{\lambda}}$ are defined in (19). Correspondingly, define the Lyapunov value prior to correction as

$$V_{k+1|k} := c_{\mathbf{x}} \|\mathbf{x}_{k+1|k} - \mathbf{x}^*(t_{k+1})\|^2 + c_{\boldsymbol{\lambda}} \|\boldsymbol{\lambda}_{k+1|k} - \boldsymbol{\lambda}^*(t_{k+1})\|^2. \quad (26)$$

The quantities $V_{k+1|k}$ and V_{k+1} measure, at time t_{k+1} , the deviation of the predicted and corrected variables from the instantaneous KKT pair. Under the step-size conditions stated in (17)–(18), the correction procedure admits the following contraction property.

Lemma 4.2 *Suppose Assumptions 2.1–2.2 hold. If the primal and dual step sizes $(\eta_{\mathbf{x}}, \eta_{\boldsymbol{\lambda}})$ satisfy (17)–(18), then the correction phase, consisting of τ primal-dual gradient steps (cf. Algorithm 1), ensures a local Q -linear contraction:*

$$V_{k+1} \leq \rho^\tau V_{k+1|k}, \quad (27)$$

where $\rho \in (0, 1)$ is the contraction factor defined in (20), and τ denotes the number of correction steps.

Proof The proof of Lemma 4.2 can be found in Appendix D. ■

This result follows from standard Lyapunov analysis for primal-dual gradient dynamics, and the contraction factor ρ matches that of the corresponding static problem in [28].

4.3 Combined Prediction-Correction Analysis

Combining the prediction error bound (24) (guaranteed by Lemma 4.1) with the τ -step contraction (27) (guaranteed by Lemma 4.2), and using the equivalence between $\sqrt{V_k}$ and $\|\mathbf{z}_k - \mathbf{z}^*(t_k)\|$ up to factors depending on $\sqrt{c_{\mathbf{x}}}$ and $\sqrt{c_{\boldsymbol{\lambda}}}$ (more details in Appendix E), we obtain the global convergence result stated in Theorem 4.3.

Theorem 4.3 *Suppose Assumptions 2.1–2.3 hold. If the step sizes $(\eta_{\mathbf{x}}, \eta_{\boldsymbol{\lambda}})$ satisfy (17)–(18) and the parameters (ρ, τ, r) satisfy (21), then the proposed PD-PC method achieves $\mathcal{O}(h^2)$ tracking accuracy. To be specific, the following statements hold:*

- 1) $\limsup_{k \rightarrow \infty} \|\mathbf{z}_k - \mathbf{z}^*(t_k)\| = \mathcal{O}(h^2),$
- 2) $\limsup_{k \rightarrow \infty} \|\nabla_{\mathbf{z}} L(\mathbf{z}_k, t_k)\| = \mathcal{O}(h^2).$

Proof The proof of Theorem 4.3 can be found in Appendix E. ■

Remark 4.4 Theorem 4.3 establishes $\mathcal{O}(h^2)$ tracking accuracy measured by the aggregate primal-dual tracking error $\|\mathbf{z}_k - \mathbf{z}^*(t_k)\|$ and the KKT residual $\|\nabla_{\mathbf{z}} L(\mathbf{z}_k, t_k)\|$. Moreover, since $\mathbf{x}_k$ is the primal block of $\mathbf{z}_k$, the APTE also satisfies $\limsup_{k \rightarrow \infty} \|\mathbf{x}_k - \mathbf{x}^*(t_k)\| = \mathcal{O}(h^2)$. This result is state-of-the-art among discrete-time, first-order (gradient-based) methods for constrained time-varying optimization. It improves upon prior approaches, including the primal-dual method in [28] and the dual-only method in [30], and matches the best-known accuracy order achieved in unconstrained settings^[6, 32]. Newton-type methods may attain higher-order accuracy (e.g., $\mathcal{O}(h^4)$)^[31], but they typically rely on second-order operations such as Hessian inversion.

5 Numerical Simulations

In this section, we validate the theoretical results and evaluate the empirical performance of the proposed PD-PC method under two representative scenarios. The first example considers time-invariant constraints $(\mathbf{A}, \mathbf{b})$, and the second explores the more general time-varying setting with $(\mathbf{A}(t), \mathbf{b}(t))$.

5.1 Example 1: Tracking Problem with Time-Invariant $(\mathbf{A}, \mathbf{b})$

We consider a d -dimensional tracking problem with a quadratic time-varying convex objective and time-invariant linear equality constraint:

$$\min_{\mathbf{x} \in \mathbb{R}^6} f(\mathbf{x}, t) := \frac{1}{2} \|\mathbf{x} - \mathbf{r}(t)\|^2 \quad \text{s.t.} \quad \mathbf{A}\mathbf{x} = \mathbf{b}, \quad (28)$$

where the constraint matrix $\mathbf{A} \in \mathbb{R}^{2 \times 6}$ has orthonormal rows (full row rank), with $\mathbf{b} = [0.3, -0.2]^\top$. The time-varying reference is defined as

$$\mathbf{r}(t) = [\sin(\omega t), \cos(\omega t), 0.5 \sin(2\omega t), 0.3 \cos(1.5\omega t), 0.2 \sin(1.2\omega t), 1]^\top, \quad (29)$$

with frequency $\omega = 0.4$. The gradient of the objective is $\nabla_{\mathbf{x}} f(\mathbf{x}, t) = \mathbf{x} - \mathbf{r}(t)$.

We compare four schemes under identical step sizes $\eta_{\mathbf{x}} = 0.2$, $\eta_{\boldsymbol{\lambda}} = 0.2$. Specifically, we consider the following: (i) The baseline Primal-Dual (PD) method^[28] without prediction, (ii) the Dual Prediction-Correction (Dual-PC) method^[30], and (iii) two instances of our proposed PD-PC method with different memory parameters $r = 1$ and $r = 5$. For the first three schemes, each predictor is followed by τ primal-dual gradient corrections (cf. (15)–(16)). In the Dual-PC scheme, the correction phase also consists of τ steps, but follows the procedure outlined in [30]. Specifically, the primal variable $\mathbf{x}$ is updated by solving the Lagrangian dual function, i.e., by computing the exact minimizer of the corresponding subproblem, while the dual variable $\boldsymbol{\lambda}$ is updated using a gradient ascent step. The initialization and predictor design for each scheme are described in detail below:

- **PD (no prediction):** Initialization $\mathbf{z}_0 = \mathbf{0}$; predictor $\mathbf{z}_{k+1|k} = \mathbf{z}_k$.
- **PD-PC ($r = 1$):** Initialization $\mathbf{z}_{-1} = \mathbf{z}_0 = \mathbf{0}$; predictor $\mathbf{z}_{k+1|k} = 2\mathbf{z}_k - \mathbf{z}_{k-1}$.
- **PD-PC ($r = 5$):** Initialization $\mathbf{z}_{-5} = \cdots = \mathbf{z}_0 = \mathbf{0}$; predictor $\mathbf{z}_{k+1|k} = \frac{6}{5}\mathbf{z}_k - \frac{1}{5}\mathbf{z}_{k-5}$.
- **Dual-PC ($P = 3$):** Initialization $\mathbf{z}_0 = \mathbf{0}$; prediction obtained by $P = 3$ and $\beta_{\text{dual}} = 0.5$ to form $(\mathbf{x}_{k+1|k}, \boldsymbol{\lambda}_{k+1|k})^\dagger$.

We evaluate the performance of all four schemes using two complementary experiments under invariant problem parameters $(\mathbf{A}, \mathbf{b})$. Figure 1 plots the time evolution of the primal tracking error $\|\mathbf{x}_k - \mathbf{x}^*(t_k)\|$ under a fixed sampling period $h = 0.01$ and increasing correction steps $\tau \in \{4, 8, 12\}$. Figure 2 fixes both the time horizon and τ sufficiently large so that all schemes converge within their asymptotic tracking regime, and then varies h to generate the APTE scaling curves.

[†]Here, P is the number of inner dual prediction steps, and β_{dual} is the corresponding step size. See [30] for full algorithmic details.

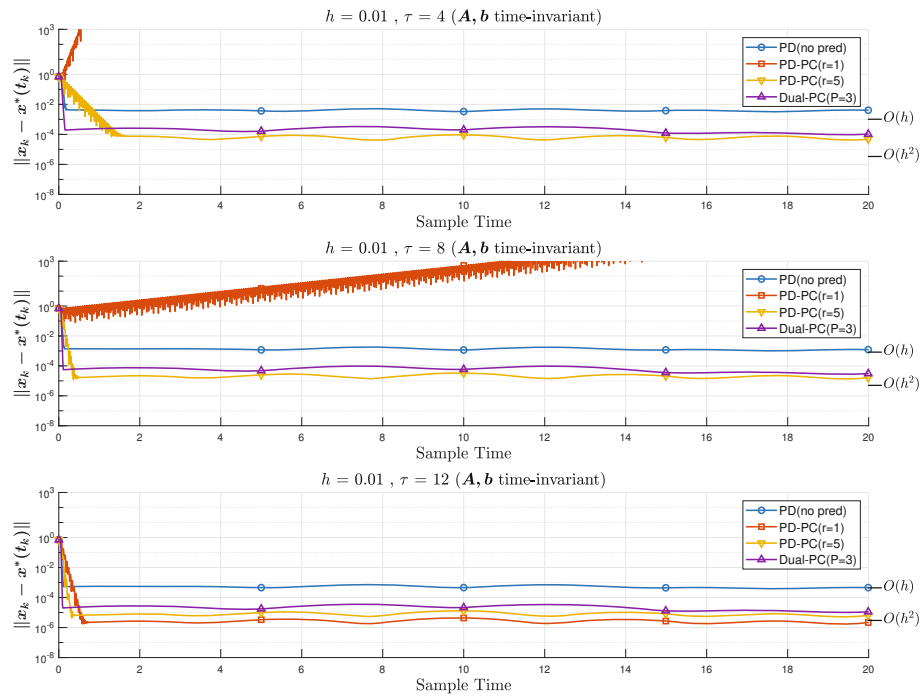


Figure 1 Primal tracking error under fixed $h = 0.01$ with increasing τ (invariant $\mathbf{A}, \mathbf{b}$)

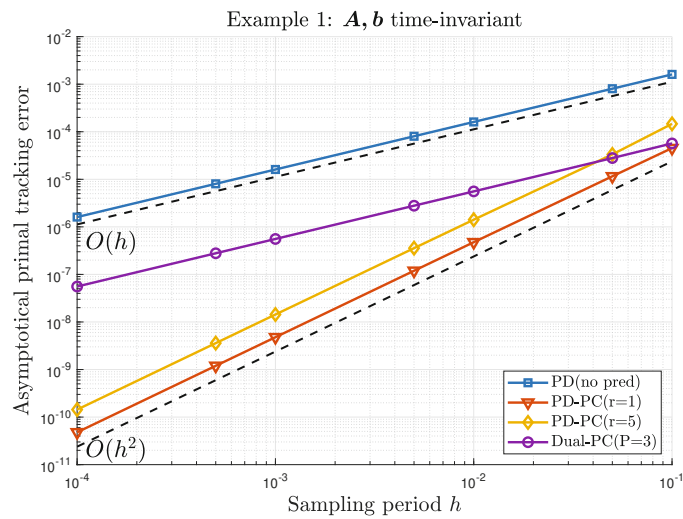


Figure 2 Asymptotic primal tracking error versus h under large τ (invariant $\mathbf{A}, \mathbf{b}$)

The observed tracking behaviors in both figures align with the theoretical results, which are summarized below.

- 1) The baseline PD method yields $O(h)$ APTE, while both PD-PC schemes (with $r = 1$ and $r = 5$) achieve $O(h^2)$ APTE. The Dual-PC method exhibits an intermediate order between

$\mathcal{O}(h)$ and $\mathcal{O}(h^2)$. These behaviors are clearly reflected in both Figures 1 and 2.

- 2) As shown in Figure 1, although PD-PC ($r = 1$) and PD-PC ($r = 5$) have the same APTE order, the $r = 5$ predictor requires fewer correction steps to converge. Specifically, PD-PC ($r = 5$) has converged at $\tau = 4$, while PD-PC ($r = 1$) still exhibits visible tracking error at $\tau = 8$, indicating that convergence has not yet been achieved. This confirms the theoretical analysis of a smaller $\tau_{\min}$ for higher r (cf. (22)).

5.2 Example 2: Tracking Problem with Time-Varying $(\mathbf{A}(t), \mathbf{b}(t))$

We now consider a more complex tracking scenario where both the objective and the feasible set evolve with time. Compared to Example 1, this setup introduces a time-varying constraint and an additional exponential penalty term in the objective. The optimization problem is formulated as

$$\min_{\mathbf{x} \in \mathbb{R}^3} f(\mathbf{x}, t) := \frac{1}{2} \|\mathbf{x} - \mathbf{r}(t)\|^2 + \mu \exp(\boldsymbol{\gamma}^\top \mathbf{x}) \quad \text{s.t.} \quad \mathbf{A}(t)\mathbf{x} = \mathbf{b}(t), \quad (30)$$

where $\mu = 0.03$ and $\boldsymbol{\gamma} = [0.6, -0.3, 0.2]^\top$ define the exponential penalty direction and weight.

The time-varying reference $\mathbf{r}(t)$ and constraint $\mathbf{A}(t)$, $\mathbf{b}(t)$ are given by

$$\mathbf{r}(t) = [\sin(\omega t), \cos(\omega t), 1]^\top, \quad \mathbf{A}(t) = [\cos(\omega t), \sin(\omega t), 1], \quad \mathbf{b}(t) = 1 + 0.2 \sin(0.7\omega t), \quad (31)$$

with frequency $\omega = 0.4$.

Unlike Example 1, the optimal trajectory $\mathbf{x}^*(t)$ here does not have a closed-form due to the exponential term and time-varying constraint $\mathbf{A}(t)\mathbf{x} = \mathbf{b}(t)$. Accordingly, we use the KKT residual $\|\nabla_{\mathbf{z}} \mathcal{L}(\mathbf{z}_k, t_k)\|$ as the performance metric, which serves as a surrogate for the APTE. Their equivalence in accuracy order is guaranteed by Theorem 4.3.

We follow the same initialization and algorithmic setup as those in Example 1, including step sizes $\eta_{\mathbf{x}} = 0.2$, $\eta_{\boldsymbol{\lambda}} = 0.2$, and the same four schemes: **PD (no prediction)**, **PD-PC** ($r = 1$), **PD-PC** ($r = 5$), and **Dual-PC** ($P = 3$)[‡]. All schemes use τ correction steps: The first three perform primal-dual gradient updates, while Dual-PC follows by solving an inner minimization for $\mathbf{x}$ and updating $\boldsymbol{\lambda}$ via gradient ascent.

We evaluate the performance of all four schemes using two complementary experiments under varying $(\mathbf{A}(t), \mathbf{b}(t))$. Figure 3 illustrates the tracking performance with a fixed stepsize $h = 0.01$ and varying correction steps $\tau \in \{4, 8, 12\}$. Figure 4 examines the asymptotic tracking behavior by sweeping over h under sufficiently large correction steps and time horizon. The only difference from Example 1 is that the KKT residual is used as the performance metric.

The results closely resemble those in Example 1: PD remains $\mathcal{O}(h)$, both PD-PC schemes achieve $\mathcal{O}(h^2)$, and Dual-PC lies in between. The $r = 5$ predictor again requires fewer correction steps than $r = 1$, in line with the theoretical reduction in $\tau_{\min}$ (cf. (22)). These results highlight the benefit of having a tunable predictor memory r , which enables a flexible trade-off between retaining more past iterates (longer memory) and performing more correction steps (higher correction cost) across distinct problem settings.

[‡]Although [30] focuses on time-invariant $(\mathbf{A}, \mathbf{b})$, we still include Dual-PC as a baseline for comparison in the time-varying setting.

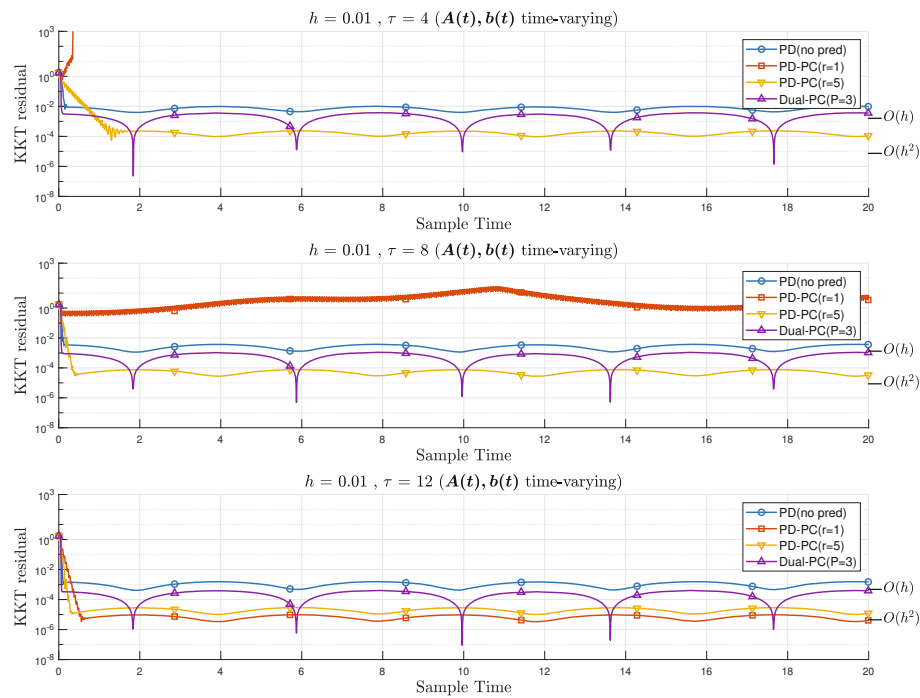


Figure 3 KKT residual under fixed $h = 0.01$ with increasing τ (time-varying $\mathbf{A}(t), \mathbf{b}(t)$)

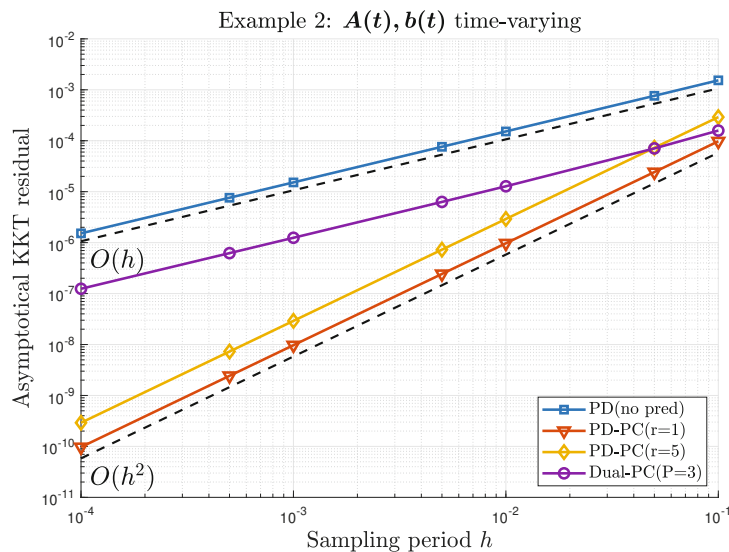


Figure 4 Asymptotic KKT residual versus h under large τ (time-varying $\mathbf{A}(t), \mathbf{b}(t)$)

Computational efficiency To complement the above tracking results, we benchmark the wall-clock time per sampled-time update for the four schemes in this example. Specifically, for each outer iteration k (updating $\mathbf{z}_k$ to $\mathbf{z}_{k+1}$), the timing includes the prediction step (if any)

and τ correction steps. All timing results are obtained using Matlab on a computer with an Intel i9-14900HX CPU (2.20 GHz) and 32 GB RAM. Table 2 reports the mean and standard deviation over 20 runs.

Table 2 Wall-clock time per sampled-time update (mean $\pm$ std, μ s/update) under different τ

Method	$\tau = 4$	$\tau = 8$	$\tau = 12$
PD (no prediction)	2.62 ± 0.11	3.84 ± 0.11	4.97 ± 0.13
PD-PC ($r = 1$)	2.94 ± 0.09	4.18 ± 0.08	5.26 ± 0.13
PD-PC ($r = 5$)	2.85 ± 0.08	4.05 ± 0.09	5.17 ± 0.12
Dual-PC ($P = 3$)	19.69 ± 0.62	32.84 ± 0.54	44.74 ± 0.74

As shown in Table 2, PD-PC incurs a wall-clock cost comparable to the baseline PD, with only a small overhead from the linear predictor. In contrast, Dual-PC is consistently about one order of magnitude slower, due to the heavier computations in its prediction/correction steps (notably the inner minimization). Overall, this benchmark suggests that PD-PC keeps a per-update runtime close to first-order primal-dual updates while achieving the improved asymptotic tracking behavior. Consequently, as the problem dimension grows and the per-update runtime becomes comparable to the sampling period h , the low-overhead PD-PC scheme becomes particularly attractive for real-time tracking.

6 Conclusions

In this work, we proposed the primal-dual prediction-correction (PD-PC) method for linearly constrained time-varying convex optimization. By combining primal-dual gradient steps with a memory-based linear predictor, the method achieves provable $\mathcal{O}(h^2)$ tracking accuracy using only first-order information. The tunable predictor enables a flexible trade-off between the amount of historical data stored and correction cost, while the correction phase remains lightweight and gradient-based. Compared to prior dual-based or second-order methods, PD-PC offers comparable accuracy with lower computational overhead. Theoretical guarantees and numerical experiments validate the method's effectiveness under both static and dynamic linear constraints.

Future work will focus on extending the PD-PC method to achieve higher tracking accuracy and distributed implementations for large-scale multi-agent systems.

Conflict of Interest

SUN Jian is an editorial board member and CHEN Jie is the Editor-in-Chief for Journal of Systems Science & Complexity, and they were not involved in the editorial review or the decision to publish this article. All authors declare that there are no competing interests.

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Appendix

A Proof of Proposition 2.4

Proof For brevity, we write $\mathbf{A}_t := \mathbf{A}(t)$ and $\mathbf{b}_t := \mathbf{b}(t)$ (and similarly $\dot{\mathbf{A}}_t, \ddot{\mathbf{A}}_t, \dot{\mathbf{b}}_t, \ddot{\mathbf{b}}_t$). Unless otherwise stated, all derivatives of f and $\mathcal{L}$ are evaluated at time t along the optimal solution, e.g., $\nabla_{\mathbf{z}\mathbf{z}}\mathcal{L}_t := \nabla_{\mathbf{z}\mathbf{z}}\mathcal{L}(\mathbf{z}^*(t), t)$ and $\nabla_{\mathbf{x}\mathbf{x}}f_t := \nabla_{\mathbf{x}\mathbf{x}}f(\mathbf{x}^*(t), t)$. The same shorthand applies throughout this proof to all mixed and higher-order derivatives.

Step 1: Uniform invertibility of the Lagrangian Hessian.

Under Assumptions 2.1–2.2, the block-structured Hessian of the Lagrangian is given by

$$\nabla_{\mathbf{z}\mathbf{z}}\mathcal{L}_t = \begin{bmatrix} \nabla_{\mathbf{x}\mathbf{x}}f_t & \mathbf{A}_t^\top \\ \mathbf{A}_t & \mathbf{0} \end{bmatrix}. \quad (\text{A.1})$$

Its Schur complement is

$$\mathbf{X}_t := (\nabla_{\mathbf{x}\mathbf{x}}f_t)^{-1}, \quad \mathbf{S}_t := \mathbf{A}_t\mathbf{X}_t\mathbf{A}_t^\top \succ 0, \quad (\text{A.2})$$

since $\nabla_{\mathbf{x}\mathbf{x}}f_t \succeq \nu\mathbf{I}$ and $\mathbf{A}_t$ has full row rank by Assumption 2.1. Hence $\nabla_{\mathbf{z}\mathbf{z}}\mathcal{L}_t$ is nonsingular and its inverse is explicitly given by

$$(\nabla_{\mathbf{z}\mathbf{z}}\mathcal{L}_t)^{-1} = \begin{bmatrix} \mathbf{X}_t - \mathbf{X}_t\mathbf{A}_t^\top\mathbf{S}_t^{-1}\mathbf{A}_t\mathbf{X}_t & \mathbf{X}_t\mathbf{A}_t^\top\mathbf{S}_t^{-1} \\ \mathbf{S}_t^{-1}\mathbf{A}_t\mathbf{X}_t & -\mathbf{S}_t^{-1} \end{bmatrix}. \quad (\text{A.3})$$

To bound $\|(\nabla_{\mathbf{z}\mathbf{z}}\mathcal{L}_t)^{-1}\|$, rewrite (A.3) as a sum of four block-sparse matrices:

$$\begin{aligned} (\nabla_{\mathbf{z}\mathbf{z}}\mathcal{L}_t)^{-1} &= \underbrace{\begin{bmatrix} \mathbf{X}_t - \mathbf{X}_t\mathbf{A}_t^\top\mathbf{S}_t^{-1}\mathbf{A}_t\mathbf{X}_t & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}}_{(a)} + \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{X}_t\mathbf{A}_t^\top\mathbf{S}_t^{-1} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}}_{(b)} \\ &\quad + \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{S}_t^{-1}\mathbf{A}_t\mathbf{X}_t & \mathbf{0} \end{bmatrix}}_{(c)} + \underbrace{\begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{S}_t^{-1} \end{bmatrix}}_{(d)}. \end{aligned} \quad (\text{A.4})$$

Applying the triangle inequality to (A.4) yields

$$\begin{aligned} \|(\nabla_{\mathbf{z}\mathbf{z}}\mathcal{L}_t)^{-1}\| &\leq \|\mathbf{X}_t - \mathbf{X}_t\mathbf{A}_t^\top\mathbf{S}_t^{-1}\mathbf{A}_t\mathbf{X}_t\| + \|\mathbf{X}_t\mathbf{A}_t^\top\mathbf{S}_t^{-1}\| + \|\mathbf{S}_t^{-1}\mathbf{A}_t\mathbf{X}_t\| + \|\mathbf{S}_t^{-1}\| \\ &\leq \|\mathbf{X}_t\| + \|\mathbf{X}_t\|\|\mathbf{A}_t\|\|\mathbf{S}_t^{-1}\|\|\mathbf{A}_t\|\|\mathbf{X}_t\| + 2\|\mathbf{X}_t\|\|\mathbf{A}_t\|\|\mathbf{S}_t^{-1}\| + \|\mathbf{S}_t^{-1}\| \\ &= \|\mathbf{X}_t\| + \|\mathbf{S}_t^{-1}\|\left(1 + \|\mathbf{A}_t\|\|\mathbf{X}_t\|\right)^2. \end{aligned} \quad (\text{A.5})$$

Invoking Assumptions 2.1 and 2.2, we obtain

$$\|\mathbf{X}_t\| = \|(\nabla_{\mathbf{x}\mathbf{x}}f_t)^{-1}\| \leq \frac{1}{\nu}, \quad \|\mathbf{A}_t\| \leq \bar{\sigma}. \quad (\text{A.6})$$

Moreover, since $\nabla_{xx}f_t \preceq \delta \mathbf{I}$, we have

$$\mathbf{S}_t = \mathbf{A}_t \mathbf{X}_t \mathbf{A}_t^T \succeq \frac{1}{\delta} \mathbf{A}_t \mathbf{A}_t^T \succeq \frac{\underline{\sigma}^2}{\delta} \mathbf{I} \implies \|\mathbf{S}_t^{-1}\| \leq \frac{\delta}{\underline{\sigma}^2}. \quad (\text{A.7})$$

Substituting bounds (A.6) and (A.7) into (A.5) yields the desired uniform bound:

$$\|(\nabla_{zz}\mathcal{L}_t)^{-1}\| \leq \frac{1}{\nu} + \frac{\delta}{\underline{\sigma}^2} \left(1 + \frac{\bar{\sigma}}{\nu}\right)^2 =: C_J. \quad (\text{A.8})$$

Step 2: Existence and boundedness of $\dot{\mathbf{z}}^*(t)$.

Under Assumptions 2.1–2.2, the primal-dual solution trajectory $\mathbf{z}^*(t)$ satisfies the differential system (9). The mixed partial $\nabla_{zt}\mathcal{L}_t$ has block form

$$\nabla_{zt}\mathcal{L}_t = \begin{bmatrix} \nabla_{xt}f_t + \dot{\mathbf{A}}_t^T \boldsymbol{\lambda}^* \\ \dot{\mathbf{A}}_t \mathbf{x}^* - \dot{\mathbf{b}}_t \end{bmatrix}. \quad (\text{A.9})$$

By Assumption 2.3 (A1)–(A2), the terms $\mathbf{x}^*$, $\boldsymbol{\lambda}^*$, $\nabla_{xt}f_t$, $\dot{\mathbf{A}}_t$, and $\dot{\mathbf{b}}_t$ are all uniformly bounded in t ; thus there exists a constant $C_t^{(1)} < \infty$ such that

$$\|\nabla_{zt}\mathcal{L}_t\| \leq C_t^{(1)}. \quad (\text{A.10})$$

Combined with the uniform bound on the inverse Hessian $\|(\nabla_{zz}\mathcal{L}_t)^{-1}\| \leq C_J$ from Step 1, we obtain

$$\|\dot{\mathbf{z}}^*(t)\| \leq C_J C_t^{(1)}, \quad \forall t \geq 0. \quad (\text{A.11})$$

Therefore, $\dot{\mathbf{z}}^*(t)$ exists and is uniformly bounded.

Step 3: Existence and boundedness of $\ddot{\mathbf{z}}^*(t)$.

Differentiating the differential system (9) in time and applying the product and chain rules gives:

$$\begin{aligned} \ddot{\mathbf{z}}^* &= -\frac{d}{dt}((\nabla_{zz}\mathcal{L}_t)^{-1}\nabla_{zt}\mathcal{L}_t) \\ &= -\left(\frac{d}{dt}(\nabla_{zz}\mathcal{L}_t)^{-1}\right)\nabla_{zt}\mathcal{L}_t - (\nabla_{zz}\mathcal{L}_t)^{-1}\left(\frac{d}{dt}\nabla_{zt}\mathcal{L}_t\right) \\ &= -\left(-(\nabla_{zz}\mathcal{L}_t)^{-1}\left(\frac{d}{dt}\nabla_{zz}\mathcal{L}_t\right)(\nabla_{zz}\mathcal{L}_t)^{-1}\right)\nabla_{zt}\mathcal{L}_t - (\nabla_{zz}\mathcal{L}_t)^{-1}\left(\frac{d}{dt}\nabla_{zt}\mathcal{L}_t\right) \\ &= (\nabla_{zz}\mathcal{L}_t)^{-1}\left(\frac{d}{dt}\nabla_{zz}\mathcal{L}_t\right)\underbrace{(\nabla_{zz}\mathcal{L}_t)^{-1}\nabla_{zt}\mathcal{L}_t}_{-\dot{\mathbf{z}}^* \text{ by (9)}} - (\nabla_{zz}\mathcal{L}_t)^{-1}\left(\frac{d}{dt}\nabla_{zt}\mathcal{L}_t\right) \\ &= -\boxed{(\nabla_{zz}\mathcal{L}_t)^{-1}\left(\frac{d}{dt}\nabla_{zz}\mathcal{L}_t\right)\dot{\mathbf{z}}^*} - (\nabla_{zz}\mathcal{L}_t)^{-1}\left(\frac{d}{dt}\nabla_{zt}\mathcal{L}_t\right). \end{aligned} \quad (\text{A.12})$$

The boxed term in (A.12) involves the total derivative of the Hessian $\nabla_{zz}\mathcal{L}_t$ along the trajectory $\mathbf{z}^*(t)$. Since $\nabla_{zz}\mathcal{L}_t$ is matrix-valued, applying the chain rule introduces the third derivative of $\mathcal{L}_t$, which can be represented as a third-order tensor (equivalently, a trilinear form). To make

the ensuing notation explicit, we follow the standard higher-order differential convention in [35]. Let $\mathcal{T}$ be a third-order tensor with entries $(\mathcal{T})_{ijk}$ and let u, v be vectors of compatible dimension. We use $\mathcal{T}[u]$ to denote the matrix with entries $(\mathcal{T}[u])_{ij} = \sum_k (\mathcal{T})_{ijk} u_k$, and $\mathcal{T}[u, v]$ to denote the vector with entries $(\mathcal{T}[u, v])_i = \sum_j \sum_k (\mathcal{T})_{ijk} u_j v_k$ (equivalently, $\mathcal{T}[u, v] = \mathcal{T}[u]v$).

With the above convention, the chain rule yields

$$\frac{d}{dt} \nabla_{zz} \mathcal{L}_t = \nabla_{zzz} \mathcal{L}_t [\dot{z}^*] + \nabla_{zzt} \mathcal{L}_t, \quad \frac{d}{dt} \nabla_{zt} \mathcal{L}_t = \nabla_{zzt} \mathcal{L}_t \dot{z}^* + \nabla_{ztt} \mathcal{L}_t. \quad (\text{A.13})$$

Substituting (A.13) into (A.12) and simplifying (by contracting the corresponding indices) gives

$$\begin{aligned} \ddot{z}^* &= -(\nabla_{zz} \mathcal{L}_t)^{-1} (\nabla_{zzz} \mathcal{L}_t [\dot{z}^*] + \nabla_{zzt} \mathcal{L}_t) \dot{z}^* - (\nabla_{zz} \mathcal{L}_t)^{-1} (\nabla_{zzt} \mathcal{L}_t \dot{z}^* + \nabla_{ztt} \mathcal{L}_t) \\ &= -(\nabla_{zz} \mathcal{L}_t)^{-1} \nabla_{zzz} \mathcal{L}_t [\dot{z}^*, \dot{z}^*] - 2(\nabla_{zz} \mathcal{L}_t)^{-1} \nabla_{zzt} \mathcal{L}_t \dot{z}^* - (\nabla_{zz} \mathcal{L}_t)^{-1} \nabla_{ztt} \mathcal{L}_t. \end{aligned} \quad (\text{A.14})$$

Since the constraint term $\lambda^T(\mathbf{A}_t \mathbf{x} - \mathbf{b}_t)$ in $\mathcal{L}_t$ is affine in $\mathbf{x}$ and linear in λ , any third-order derivative of $\mathcal{L}_t$ that involves a λ -index or differentiates $(\mathbf{A}_t \mathbf{x} - \mathbf{b}_t)$ twice with respect to $\mathbf{x}$ vanishes. Consequently, the only nonzero third-order block is $\nabla_{xxx} f_t$, which gives

$$\nabla_{zzz} \mathcal{L}_t [\dot{z}^*, \dot{z}^*] = \begin{bmatrix} \nabla_{xxx} f_t [\dot{\mathbf{x}}^*, \dot{\mathbf{x}}^*] \\ \mathbf{0} \end{bmatrix}. \quad (\text{A.15})$$

In addition, the other two derivative terms in (A.14) evaluate to

$$\nabla_{zzt} \mathcal{L}_t \dot{z}^* = \begin{bmatrix} \nabla_{xxt} f_t \dot{\mathbf{x}}^* + \dot{\mathbf{A}}_t^T \dot{\lambda}^* \\ \dot{\mathbf{A}}_t \dot{\mathbf{x}}^* \end{bmatrix}, \quad \nabla_{ztt} \mathcal{L}_t = \begin{bmatrix} \nabla_{xtt} f_t + \ddot{\mathbf{A}}_t^T \dot{\lambda}^* \\ \ddot{\mathbf{A}}_t \dot{\mathbf{x}}^* - \ddot{\mathbf{b}}_t \end{bmatrix}. \quad (\text{A.16})$$

By Assumption 2.3 (A1)–(A3), all terms on the right-hand side of (A.14) are uniformly bounded in t (since $\dot{z}^*$ is also uniformly bounded by Step 2). Thus, there exists a constant $C_t^{(2)} < \infty$ such that

$$\|\ddot{z}^*(t)\| \leq C_J C_t^{(2)}, \quad \forall t \geq 0. \quad (\text{A.17})$$

Hence, $\ddot{z}^*(t)$ exists and is uniformly bounded. ■

B Discussion of Remark 2.7

In this subsection, we justify the claim in Remark 2.7 that Assumption 2.3 (A1), i.e., the uniform boundedness of the dual trajectory $\lambda^*(t)$, can be replaced by a more natural condition on the primal problem.

The motivation is twofold: First, directly assuming $\sup_{t \geq 0} \|\lambda^*(t)\| < \infty$ is often hard to verify in practice due to the implicit definition of the dual variable; second, the boundedness of the primal objective and constraints already implies a form of regularity on the dual variables, under standard assumptions. The following lemma formalizes this claim.

Lemma B.1 Suppose Assumptions 2.1 and 2.2 hold. Assume there exist constants $G, B < \infty$ and a reference point $\mathbf{x}_{\text{ref}} \in \mathbb{R}^d$ (e.g., $\mathbf{x}_{\text{ref}} = \mathbf{0}$) such that

$$\sup_{t \geq 0} \|\nabla_{\mathbf{x}} f(\mathbf{x}_{\text{ref}}, t)\| \leq G, \quad \sup_{t \geq 0} \|\mathbf{b}(t)\| \leq B. \quad (\text{B.1})$$

Then there exist constants $G_{\mathbf{x}}, G_{\boldsymbol{\lambda}} < \infty$, independent of t , such that

$$\sup_{t \geq 0} \|\mathbf{x}^*(t)\| \leq G_{\mathbf{x}}, \quad \sup_{t \geq 0} \|\boldsymbol{\lambda}^*(t)\| \leq G_{\boldsymbol{\lambda}}. \quad (\text{B.2})$$

Hence Assumption 2.3 (A1) follows.

Proof For brevity, we write $f_t(\mathbf{x}) := f(\mathbf{x}, t)$, $\nabla f_t(\mathbf{x}) := \nabla_{\mathbf{x}} f(\mathbf{x}, t)$, $\mathbf{A}_t := \mathbf{A}(t)$, and $\mathbf{b}_t := \mathbf{b}(t)$, and denote $\mathbf{x}^*(t)$ and $\boldsymbol{\lambda}^*(t)$ by $\mathbf{x}^*$ and $\boldsymbol{\lambda}^*$ when no confusion arises.

Without loss of generality, we assume $\mathbf{x}_{\text{ref}} = \mathbf{0}$ in the sequel; otherwise, the same argument applies with additional constants depending on $\|\mathbf{x}_{\text{ref}}\|$, due to the smoothness and strong convexity of $f(\cdot, t)$.

Step 1: Bounded feasible reference solution.

Let $\mathbf{A}_t^\dagger$ denote the Moore-Penrose pseudoinverse of $\mathbf{A}_t$, which, under the full row-rank assumption $\text{rank}(\mathbf{A}_t) = m$, has the closed-form expression $\mathbf{A}_t^\dagger := \mathbf{A}_t^T (\mathbf{A}_t \mathbf{A}_t^T)^{-1}$. We define the reference solution as $\mathbf{x}_p := \mathbf{A}_t^\dagger \mathbf{b}_t$, which serves as a feasible point satisfying $\mathbf{A}_t \mathbf{x}_p = \mathbf{b}_t$. By Assumption 2.1, the smallest nonzero singular value of $\mathbf{A}_t$ is lower bounded by $\underline{\sigma}$. Hence

$$\|\mathbf{x}_p\| \leq \|\mathbf{A}_t^\dagger\| \cdot \|\mathbf{b}_t\| \leq \frac{1}{\underline{\sigma}} \cdot B =: M_p. \quad (\text{B.3})$$

Step 2: Upper bound on the optimal primal solution.

Since $\mathbf{x}^*$ minimizes Problem (1), it holds that

$$f_t(\mathbf{x}^*) \leq f_t(\mathbf{x}_p). \quad (\text{B.4})$$

On the other hand, the ν -strong convexity of $f(\cdot, t)$ yields

$$f_t(\mathbf{x}^*) \geq f_t(\mathbf{0}) + \nabla f_t(\mathbf{0})^T \mathbf{x}^* + \frac{\nu}{2} \|\mathbf{x}^*\|^2. \quad (\text{B.5})$$

Meanwhile, δ -smoothness implies

$$f_t(\mathbf{x}_p) \leq f_t(\mathbf{0}) + \nabla f_t(\mathbf{0})^T \mathbf{x}_p + \frac{\delta}{2} \|\mathbf{x}_p\|^2. \quad (\text{B.6})$$

Combining the inequalities (B.4)–(B.6) and rearranging gives

$$\frac{\nu}{2} \|\mathbf{x}^*\|^2 + \nabla f_t(\mathbf{0})^T \mathbf{x}^* \leq \nabla f_t(\mathbf{0})^T \mathbf{x}_p + \frac{\delta}{2} \|\mathbf{x}_p\|^2. \quad (\text{B.7})$$

Applying the Cauchy-Schwarz inequality on both inner products and using $\|\nabla f_t(\mathbf{0})^T\| \leq G$, $\|\mathbf{x}_p\| \leq M_p$, we obtain

$$\frac{\nu}{2} \|\mathbf{x}^*\|^2 - G \|\mathbf{x}^*\| \leq G M_p + \frac{\delta}{2} M_p^2. \quad (\text{B.8})$$

Solving this quadratic inequality gives the bound

$$\|\mathbf{x}^*\| \leq G_{\mathbf{x}} := \frac{G + \sqrt{G^2 + 2\nu \left(GM_p + \frac{\delta}{2}M_p^2\right)}}{\nu}. \quad (\text{B.9})$$

Step 3: Boundedness of the dual variable.

From the KKT condition $\nabla f_t(\mathbf{x}^*) + \mathbf{A}_t^T \boldsymbol{\lambda}^* = \mathbf{0}$, we obtain

$$\|\mathbf{A}_t^T \boldsymbol{\lambda}^*\| = \|\nabla f_t(\mathbf{x}^*)\| \leq \|\nabla f_t(\mathbf{0})\| + \delta \|\mathbf{x}^*\| \leq G + \delta G_{\mathbf{x}}, \quad (\text{B.10})$$

where the last step uses smoothness and the bound (B.9). Since $\sigma_{\min}^+(\mathbf{A}_t) \geq \underline{\sigma}$, we have

$$\|\boldsymbol{\lambda}^*\| \leq G_{\boldsymbol{\lambda}} := \frac{G + \delta G_{\mathbf{x}}}{\underline{\sigma}}. \quad (\text{B.11})$$

The bounds (B.9) and (B.11) complete the proof. $\blacksquare$

This lemma establishes that, provided the standard conditions of strong convexity, smoothness, and full row-rank constraints are satisfied, the uniform boundedness of both the primal and dual optimal trajectories follows directly from a simple upper bound on the gradient at a fixed reference point and $\mathbf{b}(t)$. Consequently, the less intuitive requirement $\sup_t \|\boldsymbol{\lambda}^*(t)\| < \infty$ in Assumption 2.3 (A1) can be replaced by a more transparent, and practically verifiable condition: $\sup_t \|\nabla_{\mathbf{x}} f(\mathbf{x}_{\text{ref}}, t)\| < \infty$.

C Proof of Lemma 4.1

Proof We prove Lemma 4.1 by performing second-order Taylor expansions of the primal-dual solution trajectory $\mathbf{z}^*(t)$ at time t_{k+1} from two directions. By Proposition 2.4 and Assumptions 2.1–2.3, the trajectory $\mathbf{z}^*(t)$ is continuously differentiable on $\mathbb{R}_+$ with uniformly bounded first and second derivatives.

By Taylor's theorem with Lagrange remainder, the trajectory $\mathbf{z}^*(t_k)$ can be expanded around t_{k+1} as

$$\mathbf{z}^*(t_k) = \mathbf{z}^*(t_{k+1}) - h\dot{\mathbf{z}}^*(t_{k+1}) + \frac{h^2}{2}\ddot{\mathbf{z}}^*(\xi_k), \quad (\text{C.1})$$

for some $\xi_k \in [t_k, t_{k+1}]$. Similarly, the expansion of $\mathbf{z}^*(t_{k-r})$ about t_{k+1} yields

$$\mathbf{z}^*(t_{k-r}) = \mathbf{z}^*(t_{k+1}) - (r+1)h\dot{\mathbf{z}}^*(t_{k+1}) + \frac{((r+1)h)^2}{2}\ddot{\mathbf{z}}^*(\xi_{k,r}), \quad (\text{C.2})$$

for some $\xi_{k,r} \in [t_{k-r}, t_{k+1}]$. Taking a linear combination of (C.1) and (C.2) gives

$$\begin{aligned} & \frac{r+1}{r}\mathbf{z}^*(t_k) - \frac{1}{r}\mathbf{z}^*(t_{k-r}) \\ &= \frac{r+1}{r} \left[\mathbf{z}^*(t_{k+1}) - h\dot{\mathbf{z}}^*(t_{k+1}) + \frac{h^2}{2}\ddot{\mathbf{z}}^*(\xi_k) \right] \\ & \quad - \frac{1}{r} \left[\mathbf{z}^*(t_{k+1}) - (r+1)h\dot{\mathbf{z}}^*(t_{k+1}) + \frac{((r+1)h)^2}{2}\ddot{\mathbf{z}}^*(\xi_{k,r}) \right] \\ &= \mathbf{z}^*(t_{k+1}) + \frac{r+1}{r} \cdot \frac{h^2}{2}\ddot{\mathbf{z}}^*(\xi_k) - \frac{1}{r} \cdot \frac{((r+1)h)^2}{2}\ddot{\mathbf{z}}^*(\xi_{k,r}). \end{aligned} \quad (\text{C.3})$$

Due to the boundedness of $\ddot{z}^*(t)$, the remainder term satisfies

$$\left\| \frac{r+1}{r} \cdot \frac{h^2}{2} \ddot{z}^*(\xi_k) - \frac{1}{r} \cdot \frac{((r+1)h)^2}{2} \ddot{z}^*(\xi_{k,r}) \right\| \leq \varepsilon h^2, \quad (\text{C.4})$$

for some constant $\varepsilon > 0$ depending on the bound in Assumption 2.3.

Combining (C.3) and (C.4), we obtain

$$z^*(t_{k+1}) = \frac{r+1}{r} z^*(t_k) - \frac{1}{r} z^*(t_{k-r}) + \mathcal{O}(h^2), \quad (\text{C.5})$$

which completes the proof. $\blacksquare$

D Proof of Lemma 4.2

Proof We prove that the correction phase in Algorithm 1 achieves a local Q -linear contraction under the step-size conditions (17)–(18).

Step 1: Setup and error dynamics.

The analysis below focuses on the correction phase at time t_{k+1} , where all quantities are evaluated accordingly. To simplify notation, denote the KKT pair at time t_{k+1} by $(\mathbf{x}^*, \boldsymbol{\lambda}^*) := (\mathbf{x}^*(t_{k+1}), \boldsymbol{\lambda}^*(t_{k+1}))$, and define $f_{k+1}(\cdot) := f(\cdot, t_{k+1})$, $\nabla f_{k+1}(\cdot) := \nabla_{\mathbf{x}} f(\cdot, t_{k+1})$, $\mathbf{A}_{k+1} := \mathbf{A}(t_{k+1})$, and $\mathbf{b}_{k+1} := \mathbf{b}(t_{k+1})$.

The correction procedure follows the primal-dual iterations in (15)–(16), initialized at the predicted point $(\mathbf{x}_{k+1|k}, \boldsymbol{\lambda}_{k+1|k})$. Let us denote the deviations from the KKT pair as

$$\tilde{\mathbf{x}}^{(s)} := \hat{\mathbf{x}}_{k+1}^{(s)} - \mathbf{x}^*, \quad \tilde{\boldsymbol{\lambda}}^{(s)} := \hat{\boldsymbol{\lambda}}_{k+1}^{(s)} - \boldsymbol{\lambda}^*. \quad (\text{D.1})$$

By subtracting the KKT relations $\nabla f_{k+1}(\mathbf{x}^*) + \mathbf{A}_{k+1}^T \boldsymbol{\lambda}^* = 0$ and $\mathbf{A}_{k+1} \mathbf{x}^* = \mathbf{b}_{k+1}$ from the updates, we obtain the coupled error system:

$$\tilde{\mathbf{x}}^{(s+1)} = \tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \left(\nabla f_{k+1}(\mathbf{x}^* + \tilde{\mathbf{x}}^{(s)}) - \nabla f_{k+1}(\mathbf{x}^*) + \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)} \right), \quad (\text{D.2})$$

$$\tilde{\boldsymbol{\lambda}}^{(s+1)} = \tilde{\boldsymbol{\lambda}}^{(s)} + \eta_{\boldsymbol{\lambda}} \mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s+1)}. \quad (\text{D.3})$$

Step 2: Auxiliary Lyapunov function candidate and recursive bound.

To analyze the behavior of the correction iterates, we define an auxiliary Lyapunov function candidate at each intermediate step s :

$$V_{k+1}^{(s)} := c_{\mathbf{x}} \|\tilde{\mathbf{x}}^{(s)}\|^2 + c_{\boldsymbol{\lambda}} \|\tilde{\boldsymbol{\lambda}}^{(s)}\|^2, \quad (\text{D.4})$$

where the weighting coefficients $c_{\mathbf{x}}, c_{\boldsymbol{\lambda}}$ are as in (19). Note that this generalizes the expressions in (25) and (26):

$$V_{k+1}^{(0)} = V_{k+1|k}, \quad V_{k+1}^{(\tau)} = V_{k+1}. \quad (\text{D.5})$$

This notation allows us to express the contraction property across any individual correction step. Specifically, we will show that for each step of the correction phase,

$$V_{k+1}^{(s+1)} \leq \rho V_{k+1}^{(s)}, \quad (\text{D.6})$$

and after performing τ correction steps, $V_{k+1} \leq \rho^\tau V_{k+1|k}$ holds, thereby establishing Lemma 4.2.

For brevity, define the intermediate gradient difference:

$$\Delta^{(s)} := \nabla f_{k+1}(\mathbf{x}^* + \tilde{\mathbf{x}}^{(s)}) - \nabla f_{k+1}(\mathbf{x}^*), \quad (\text{D.7})$$

which allows us to rewrite the update in (D.2) as

$$\tilde{\mathbf{x}}^{(s+1)} = \tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \left(\Delta^{(s)} + \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)} \right). \quad (\text{D.8})$$

Taking the norm of (D.8) gives

$$\|\tilde{\mathbf{x}}^{(s+1)}\|^2 = \|\tilde{\mathbf{x}}^{(s)}\|^2 - 2\eta_{\mathbf{x}} \langle \tilde{\mathbf{x}}^{(s)}, \Delta^{(s)} + \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)} \rangle + \eta_{\mathbf{x}}^2 \|\Delta^{(s)} + \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}\|^2. \quad (\text{D.9})$$

A similar expansion follows from (D.3):

$$\|\tilde{\boldsymbol{\lambda}}^{(s+1)}\|^2 = \|\tilde{\boldsymbol{\lambda}}^{(s)}\|^2 + 2\eta_{\boldsymbol{\lambda}} \langle \tilde{\boldsymbol{\lambda}}^{(s)}, \mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s+1)} \rangle + \eta_{\boldsymbol{\lambda}}^2 \|\mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s+1)}\|^2. \quad (\text{D.10})$$

Multiplying (D.10) by $c_{\boldsymbol{\lambda}} = \eta_{\mathbf{x}}/\eta_{\boldsymbol{\lambda}}$ and add it to (D.9):

$$\begin{aligned} & \|\tilde{\mathbf{x}}^{(s+1)}\|^2 + c_{\boldsymbol{\lambda}} \|\tilde{\boldsymbol{\lambda}}^{(s+1)}\|^2 \\ &= \|\tilde{\mathbf{x}}^{(s)}\|^2 - 2\eta_{\mathbf{x}} \langle \tilde{\mathbf{x}}^{(s)}, \Delta^{(s)} + \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)} \rangle + \eta_{\mathbf{x}}^2 \|\Delta^{(s)} + \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}\|^2 \\ & \quad + c_{\boldsymbol{\lambda}} \|\tilde{\boldsymbol{\lambda}}^{(s)}\|^2 + 2\eta_{\mathbf{x}} \langle \tilde{\boldsymbol{\lambda}}^{(s)}, \mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s+1)} \rangle + \eta_{\mathbf{x}} \eta_{\boldsymbol{\lambda}} \|\mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s+1)}\|^2. \end{aligned} \quad (\text{D.11})$$

Use the primal update $\tilde{\mathbf{x}}^{(s+1)} = \tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} (\Delta^{(s)} + \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)})$ to rewrite

$$\mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s+1)} = \mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \mathbf{A}_{k+1} \Delta^{(s)} - \eta_{\mathbf{x}} \mathbf{A}_{k+1} \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}. \quad (\text{D.12})$$

Then the mixed inner products cancel exactly:

$$\begin{aligned} & -2\eta_{\mathbf{x}} \langle \tilde{\mathbf{x}}^{(s)}, \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)} \rangle + 2\eta_{\mathbf{x}} \langle \tilde{\boldsymbol{\lambda}}^{(s)}, \mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s+1)} \rangle \\ &= -2\eta_{\mathbf{x}} \langle \tilde{\mathbf{x}}^{(s)}, \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)} \rangle + 2\eta_{\mathbf{x}} \langle \tilde{\boldsymbol{\lambda}}^{(s)}, \mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \mathbf{A}_{k+1} \Delta^{(s)} - \eta_{\mathbf{x}} \mathbf{A}_{k+1} \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)} \rangle \\ &= -2\eta_{\mathbf{x}}^2 \langle \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}, \Delta^{(s)} \rangle - 2\eta_{\mathbf{x}}^2 \|\mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}\|^2, \end{aligned} \quad (\text{D.13})$$

where $\langle \tilde{\boldsymbol{\lambda}}^{(s)}, \mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s)} \rangle = \langle \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}, \tilde{\mathbf{x}}^{(s)} \rangle$ is used. Substituting (D.13) and (D.12) into (D.11), and expanding $\|\Delta^{(s)} + \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}\|^2 = \|\Delta^{(s)}\|^2 + 2\langle \Delta^{(s)}, \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)} \rangle + \|\mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}\|^2$, the cross term $2\eta_{\mathbf{x}}^2 \langle \Delta^{(s)}, \mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)} \rangle$ cancels with the first term on the right-hand side of (D.13), and we obtain

$$\begin{aligned} & \|\tilde{\mathbf{x}}^{(s+1)}\|^2 + c_{\boldsymbol{\lambda}} \|\tilde{\boldsymbol{\lambda}}^{(s+1)}\|^2 \\ & \leq \|\tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \Delta^{(s)}\|^2 + c_{\boldsymbol{\lambda}} \|\tilde{\boldsymbol{\lambda}}^{(s)}\|^2 - \eta_{\mathbf{x}}^2 \|\mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}\|^2 + \eta_{\mathbf{x}} \eta_{\boldsymbol{\lambda}} \|\mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s+1)}\|^2. \end{aligned} \quad (\text{D.14})$$

Use $\|\mathbf{A}_{k+1} \tilde{\mathbf{x}}^{(s+1)}\| \leq \bar{\sigma} \|\tilde{\mathbf{x}}^{(s+1)}\|$ to upper bound the last term in (D.14) by $\eta_{\mathbf{x}} \eta_{\boldsymbol{\lambda}} \bar{\sigma}^2 \|\tilde{\mathbf{x}}^{(s+1)}\|^2$, and move it to the left:

$$\begin{aligned} & \underbrace{(1 - \eta_{\mathbf{x}} \eta_{\boldsymbol{\lambda}} \bar{\sigma}^2)}_{= c_{\mathbf{x}}} \|\tilde{\mathbf{x}}^{(s+1)}\|^2 + c_{\boldsymbol{\lambda}} \|\tilde{\boldsymbol{\lambda}}^{(s+1)}\|^2 \\ & \leq \|\tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \Delta^{(s)}\|^2 + c_{\boldsymbol{\lambda}} \|\tilde{\boldsymbol{\lambda}}^{(s)}\|^2 - \eta_{\mathbf{x}}^2 \|\mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}\|^2. \end{aligned} \quad (\text{D.15})$$

Finally, using $\|\mathbf{A}_{k+1}^T \tilde{\boldsymbol{\lambda}}^{(s)}\| \geq \underline{c} \|\tilde{\boldsymbol{\lambda}}^{(s)}\|$ gives

$$\begin{aligned} V_{k+1}^{(s+1)} &\leq \|\tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \Delta^{(s)}\|^2 + (c_{\boldsymbol{\lambda}} - \eta_{\mathbf{x}}^2 \underline{c}^2) \|\tilde{\boldsymbol{\lambda}}^{(s)}\|^2 \\ &= \|\tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \Delta^{(s)}\|^2 + (1 - \eta_{\mathbf{x}} \eta_{\boldsymbol{\lambda}} \underline{c}^2) c_{\boldsymbol{\lambda}} \|\tilde{\boldsymbol{\lambda}}^{(s)}\|^2. \end{aligned} \quad (\text{D.16})$$

Step 3: Bounding the primal descent term.

Note that

$$\|\tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \Delta^{(s)}\|^2 = \|\tilde{\mathbf{x}}^{(s)}\|^2 - 2\eta_{\mathbf{x}} \langle \tilde{\mathbf{x}}^{(s)}, \Delta^{(s)} \rangle + \eta_{\mathbf{x}}^2 \|\Delta^{(s)}\|^2. \quad (\text{D.17})$$

The function $f_{k+1}(\cdot)$ is ν -strongly convex and δ -smooth, which implies that

$$\langle \tilde{\mathbf{x}}^{(s)}, \Delta^{(s)} \rangle \geq \nu \|\tilde{\mathbf{x}}^{(s)}\|^2, \quad \|\Delta^{(s)}\|^2 \leq \delta \langle \tilde{\mathbf{x}}^{(s)}, \Delta^{(s)} \rangle. \quad (\text{D.18})$$

Substitute (D.18) into (D.17) to obtain

$$\begin{aligned} \|\tilde{\mathbf{x}}^{(s)} - \eta_{\mathbf{x}} \Delta^{(s)}\|^2 &\leq \|\tilde{\mathbf{x}}^{(s)}\|^2 - \eta_{\mathbf{x}} (2 - \eta_{\mathbf{x}} \delta) \langle \tilde{\mathbf{x}}^{(s)}, \Delta^{(s)} \rangle \\ &\leq (1 - \eta_{\mathbf{x}} \nu (2 - \eta_{\mathbf{x}} \delta)) \|\tilde{\mathbf{x}}^{(s)}\|^2. \end{aligned} \quad (\text{D.19})$$

Define $\rho_1 := 1 - \eta_{\mathbf{x}} \nu (1 - \eta_{\mathbf{x}} \delta)$. Then we can rewrite (D.19) as

$$\begin{aligned} (1 - \eta_{\mathbf{x}} \nu (2 - \eta_{\mathbf{x}} \delta)) \|\tilde{\mathbf{x}}^{(s)}\|^2 &= \rho_1 \|\tilde{\mathbf{x}}^{(s)}\|^2 - \eta_{\mathbf{x}} \nu \|\tilde{\mathbf{x}}^{(s)}\|^2 \\ &\leq \rho_1 c_{\mathbf{x}} \|\tilde{\mathbf{x}}^{(s)}\|^2, \end{aligned} \quad (\text{D.20})$$

where the last inequality follows from $c_{\mathbf{x}} = 1 - \eta_{\mathbf{x}} \eta_{\boldsymbol{\lambda}} \bar{\sigma}^2$ and the step-size condition $\eta_{\boldsymbol{\lambda}} \leq \nu / \bar{\sigma}^2$, which together ensure $\rho_1 (1 - c_{\mathbf{x}}) - \eta_{\mathbf{x}} \nu \leq 0$.

Similarly, define $\rho_2 := 1 - \eta_{\mathbf{x}} \eta_{\boldsymbol{\lambda}} \underline{c}^2$. Combining (D.16) and (D.20) gives

$$\begin{aligned} V_{k+1}^{(s+1)} &\leq \rho_1 c_{\mathbf{x}} \|\tilde{\mathbf{x}}^{(s)}\|^2 + \rho_2 c_{\boldsymbol{\lambda}} \|\tilde{\boldsymbol{\lambda}}^{(s)}\|^2 \\ &\leq \max\{\rho_1, \rho_2\} V_{k+1}^{(s)} \\ &= \rho V_{k+1}^{(s)}, \quad \rho \in (0, 1), \end{aligned} \quad (\text{D.21})$$

where ρ is defined in (20).

Step 4: Concluding the contraction.

Applying (D.21) recursively over τ correction steps yields

$$V_{k+1} = V_{k+1}^{(\tau)} \leq \rho^{\tau} V_{k+1}^{(0)} = \rho^{\tau} V_{k+1|k}, \quad (\text{D.22})$$

which establishes the contraction property (27) in Lemma 4.2. ■

E Proof of Theorem 4.3

Proof Under (17)–(18), one has $c_{\mathbf{x}}, c_{\boldsymbol{\lambda}} > 0$ by (19). Recalling the Lyapunov function V_k defined in (25), we have the norm equivalence

$$\min\{c_{\mathbf{x}}, c_{\boldsymbol{\lambda}}\} \|\mathbf{z}_k - \mathbf{z}^*(t_k)\|^2 \leq V_k \leq \max\{c_{\mathbf{x}}, c_{\boldsymbol{\lambda}}\} \|\mathbf{z}_k - \mathbf{z}^*(t_k)\|^2, \quad (\text{E.1})$$

which implies

$$\begin{aligned}\sqrt{V_k} &\leq \sqrt{\max\{c_x, c_\lambda\}} \|z_k - z^*(t_k)\|, \\ \|z_k - z^*(t_k)\| &\leq \frac{1}{\sqrt{\min\{c_x, c_\lambda\}}} \sqrt{V_k}.\end{aligned}\quad (\text{E.2})$$

From the prediction error bound (24),

$$\|z_{k+1|k} - z^*(t_{k+1})\| \leq \frac{r+1}{r} \|z_k - z^*(t_k)\| + \frac{1}{r} \|z_{k-r} - z^*(t_{k-r})\| + \varepsilon h^2,$$

and using (E.2) to pass between $\|\cdot\|$ and $\sqrt{V}$, we obtain

$$\begin{aligned}\sqrt{V_{k+1|k}} &\leq \sqrt{\max\{c_x, c_\lambda\}} \left(\frac{r+1}{r} \|z_k - z^*(t_k)\| + \frac{1}{r} \|z_{k-r} - z^*(t_{k-r})\| \right) + \sqrt{\max\{c_x, c_\lambda\}} \varepsilon h^2 \\ &\leq \sqrt{\frac{\max\{c_x, c_\lambda\}}{\min\{c_x, c_\lambda\}}} \left(\frac{r+1}{r} \sqrt{V_k} + \frac{1}{r} \sqrt{V_{k-r}} \right) + \sqrt{\max\{c_x, c_\lambda\}} \varepsilon h^2.\end{aligned}\quad (\text{E.3})$$

Applying the τ -step contraction (27) and taking square-roots gives

$$\sqrt{V_{k+1}} \leq \rho^{\tau/2} \sqrt{V_{k+1|k}}. \quad (\text{E.4})$$

Combining (E.3) and (E.4), and using $ax + by \leq (a+b) \max\{x, y\}$ with $a = (r+1)/r$, $b = 1/r$, yields

$$\sqrt{V_{k+1}} \leq \underbrace{\rho^{\tau/2} \sqrt{\frac{\max\{c_x, c_\lambda\}}{\min\{c_x, c_\lambda\}}} \left(\frac{r+2}{r} \right) \max\{\sqrt{V_k}, \sqrt{V_{k-r}}\}}_{:=\theta} + \underbrace{\rho^{\tau/2} \sqrt{\max\{c_x, c_\lambda\}} \varepsilon h^2}_{:=C}. \quad (\text{E.5})$$

By the condition (21), we have $\theta^2 < 1$, so $\theta \in (0, 1)$.

Starting from (E.5), we apply the inequality recursively. At each step, the maximum term $\max\{\sqrt{V_k}, \sqrt{V_{k-r}}\}$ is contracted by at most a factor θ and accumulates an additive bias of at most Ch^2 . Since the index in the recurrence decreases by at least 1 (e.g., from $k+1$ to k) and at most $r+1$ (e.g., from $k+1$ to $k-r$) in each step, the number of recursive steps required to go back from k to the initial window $\{-r, \dots, 0\}$ is at least $\lceil k/(r+1) \rceil$ and at most k . Hence, the contraction factor θ is applied at least $\lceil k/(r+1) \rceil$ times and at most k times, leading to a bound on the leading term of $\theta^{\lceil k/(r+1) \rceil} \max_{-r \leq j \leq 0} \sqrt{V_j}$. The additive bias accumulates as a geometric series with ratio θ , resulting in a total contribution bounded by $\sum_{i=0}^k \theta^i \cdot Ch^2 < \frac{C}{1-\theta} h^2$. Therefore, unrolling (E.5) gives

$$\sqrt{V_k} < \theta^{\lceil \frac{k}{r+1} \rceil} \max_{-r \leq j \leq 0} \sqrt{V_j} + \frac{C}{1-\theta} h^2, \quad (\text{E.6})$$

so that $\limsup_{k \rightarrow \infty} \sqrt{V_k} = \mathcal{O}(h^2)$.

For Statement 1, note that

$$\|z_k - z^*(t_k)\| \leq \sqrt{\frac{V_k}{\min\{c_x, c_\lambda\}}}. \quad (\text{E.7})$$

Hence, we obtain $\limsup_{k \rightarrow \infty} \|\mathbf{z}_k - \mathbf{z}^*(t_k)\| = \mathcal{O}(h^2)$, which proves Statement 1.

For Statement 2, using the definition of $\|\nabla_{\mathbf{z}} \mathcal{L}(\mathbf{z}_k, t_k)\|$ and the triangle inequality, we obtain

$$\|\nabla_{\mathbf{z}} \mathcal{L}(\mathbf{z}_k, t_k)\| \leq \|\nabla_{\mathbf{x}} f(\mathbf{x}_k, t_k) + \mathbf{A}(t_k)^T \boldsymbol{\lambda}_k\| + \|\mathbf{A}(t_k) \mathbf{x}_k - \mathbf{b}(t_k)\|, \quad (\text{E.8})$$

and since $\nabla_{\mathbf{x}} f(\mathbf{x}^*(t_k), t_k) + \mathbf{A}(t_k)^T \boldsymbol{\lambda}^*(t_k) = \mathbf{0}$ and $\mathbf{A}(t_k) \mathbf{x}^*(t_k) - \mathbf{b}(t_k) = \mathbf{0}$,

$$\begin{aligned} & \|\nabla_{\mathbf{x}} f(\mathbf{x}_k, t_k) + \mathbf{A}(t_k)^T \boldsymbol{\lambda}_k\| \\ &= \|[\nabla_{\mathbf{x}} f(\mathbf{x}_k, t_k) - \nabla_{\mathbf{x}} f(\mathbf{x}^*(t_k), t_k)] + \mathbf{A}(t_k)^T (\boldsymbol{\lambda}_k - \boldsymbol{\lambda}^*(t_k))\| \end{aligned} \quad (\text{E.9})$$

$$\begin{aligned} &\leq \delta \|\mathbf{x}_k - \mathbf{x}^*(t_k)\| + \bar{\sigma} \|\boldsymbol{\lambda}_k - \boldsymbol{\lambda}^*(t_k)\|, \\ &\|\mathbf{A}(t_k) \mathbf{x}_k - \mathbf{b}(t_k)\| = \|\mathbf{A}(t_k)(\mathbf{x}_k - \mathbf{x}^*(t_k))\| \leq \bar{\sigma} \|\mathbf{x}_k - \mathbf{x}^*(t_k)\|. \end{aligned} \quad (\text{E.10})$$

Therefore,

$$\|\nabla_{\mathbf{z}} \mathcal{L}(\mathbf{z}_k, t_k)\| \leq (\delta + \bar{\sigma}) \|\mathbf{x}_k - \mathbf{x}^*(t_k)\| + \bar{\sigma} \|\boldsymbol{\lambda}_k - \boldsymbol{\lambda}^*(t_k)\| \leq (\delta + 2\bar{\sigma}) \|\mathbf{z}_k - \mathbf{z}^*(t_k)\|, \quad (\text{E.11})$$

where the last step uses $\|\mathbf{x}_k - \mathbf{x}^*(t_k)\| \leq \|\mathbf{z}_k - \mathbf{z}^*(t_k)\|$ and $\|\boldsymbol{\lambda}_k - \boldsymbol{\lambda}^*(t_k)\| \leq \|\mathbf{z}_k - \mathbf{z}^*(t_k)\|$. By the convergence bound (E.7), we conclude

$$\limsup_{k \rightarrow \infty} \|\nabla_{\mathbf{z}} \mathcal{L}(\mathbf{z}_k, t_k)\| = \mathcal{O}(h^2), \quad (\text{E.12})$$

which proves Statement 2.

Combining the above results, we obtain both statements of Theorem 4.3, and the proof is completed. $\blacksquare$